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ABSTRACT

The coexistence and interaction of shock waves and turbulence occur in various applications, such as inertial confinement fusion, scramjet propulsion, and supernova explosions. The supersonic Taylor–Green vortex (TGV) flow is a benchmark example for studying shock/turbulence interaction problems. In this paper, the performance of the interpolation-based weighted essentially non-oscillatory (WENO) schemes for compressible TGV simulation was evaluated, rather than the existing reconstruction-based approach. First, based on the popular unequal-sized WENO (US-WENO) scheme, we developed an interpolation-based US-WENO scheme for simulating three-dimensional inviscid/viscous TGV problems over the Mach number range of 0.1–2.5. Second, a discontinuous sensor based on extremum properties (EPs) of the polynomial was designed, and a corresponding hybrid interpolation-based US-WENO scheme was developed. This EP-based sensor does not contain empirical parameters and can simulate the supersonic TGV problems well while significantly improving the computational efficiency of the original US-WENO scheme. Numerical experiments show that the interpolation-based WENO scheme has smaller numerical dissipation and better performance for compressible TGV problems than the reconstruction-based WENO scheme. However, its computational cost is slightly higher, while the hybrid US-WENO scheme can perform better in terms of both computational accuracy and efficiency.

I. INTRODUCTION

Turbulence is a complex flow phenomenon that widely exists in nature and engineering fields and controls many physical processes of numerous transient flows with practical significance, e.g., material mixing, flow separation, combustion, and astrophysics.[9,16,24,25] So, it has important scientific value and engineering significance for research. Due to the complexity of turbulence, there is no analytical solution. Therefore, numerical simulations, such as the direct numerical simulation (DNS) method that can resolve all dynamic scales in the flow field and the large eddy simulation (LES) method that models viscous dissipation at small scales, have become important means of turbulence research at the present stage. However, the inherent numerical dissipation of numerical methods has a significant impact on the accuracy and computational cost of turbulence simulation, making verification and validation key steps in developing any numerical method.

Taylor–Green vortex (TGV) flow involves the transition from laminar to turbulent regime of decaying vortices, while the fluid's relative motion leads to the complex phenomenon of generation, merging, separation, and reconnection of thick vortical structures. These make it a benchmark example widely used in turbulence research and numerical algorithm validation. In 1983, Brachet *et al.*[4] investigated the dynamics of both inviscid and viscous three-dimensional TGV flows with multiple Reynolds numbers by direct spectral numerical solution of the Navier–Stokes equations. In Ref. 9, Drikakis *et al.* tested the performance of conventional LES and monotone integrated LES in emulating the dynamics of transition to turbulence in TGV flows. For incompressible TGV flow, Yang and Pullin[44] developed a theoretical framework and a numerical method for computing the evolution of a vortex-surface field. Sharma and Sengupta[30] analyzed the vorticity dynamics of three-dimensional TGV flows, and the study showed that numerical errors produce period-doubling bifurcations, which lead to new spatial symmetries remaining unchanged at intermediate times, followed by simultaneous stretching and breaking of vortices resulting in a decaying turbulence flow. In 2021, Pereira *et al.*[24] used the partially averaged Navier–Stokes equations model to predict the transitional TGV flow at Reynolds number 3000. In Ref. 27, with the rise of machine learning methods, gene expression programing was introduced to train LES models for simulating TGV flows. For TGV flow in the field of combustion, Abdelsamie *et al.*[1] evaluated and compared a hydrogen–oxygen flame interacting with the TGV using three open-source low-Mach number DNS solvers. Similar work was done by Xu and Chen in Ref. 43, where they performed DNSs to study the dynamics of a hydrogen diffusion flame embedded in the TGV flow. In addition, the TGV problem is widely used to verify the performance of newly developed numerical algorithms, such as wave appropriate discontinuity sensor approach,[6] discrete unified gas kinetic scheme,[12,26] smoothed particle hydrodynamics method,[29] and semi-Lagrangian lattice Boltzmann method.[41]

High-order methods have become a more attractive tool for simulating TGV problems due to their smaller numerical dissipation and dispersion. Therefore, many scholars have developed high-order methods in recent years and verified their performance through TGV flow. Examples include weighted essentially non-oscillatory (WENO) schemes,[7,19,23,33] multi-resolution WENO (MR-WENO) schemes,[35,37] Targeted ENO (TENO) schemes,[7,19] discontinuous Galerkin methods,[7,11,20] spectral method,[7,33] flux reconstruction methods,[5,10] gas-kinetic scheme,[34] two-stage fourth-order subcell finite volume method,[45] and weighted compact nonlinear scheme.[47] Among them, the finite difference WENO-type scheme is more popular due to its computational efficiency and stability. The milestone work of the WENO scheme originated from the work of Jiang and Shu in Ref. 14, commonly referred to as the WENO-JS scheme. Subsequently, various variants were derived based on this, such as WENO-M,[13] WENO-Z,[3] unequal-sized WENO (US-WENO),[48,50] MR-WENO,[36,52] and others.[2,42] For its development, please refer to Shu's recent review.[32]

Specifically, one fact is that early literature focused more on high-order schemes for incompressible or weakly compressible TGV problems, but recently compressible TGV flow has begun to receive more attention. Different from subsonic TGV flow, as the Mach number increases, the compression effect is gradually enhanced, resulting in many moving shocklets and interaction with vortices. Therefore, it is more suitable for testing the performance of shock-capturing schemes and also poses greater challenges to the numerical dissipation and stability of developed numerical algorithms. Peng and Yang[23] investigated the evolution of vortex-surface fields in compressible Taylor–Green flows at Mach numbers ranging from 0.5 to 2.0 by the hybrid compact eighth-order finite difference and seventh-order WENO scheme. In 2021, Lusher and Sandham[19] first extended classical subsonic TGV flows to Mach 1.25 with Reynolds number 1600 and evaluated the effectiveness of a series of WENO/TENO (including their hybrid versions) shock-capturing schemes. Other follow-up work can be found in Refs. 7, 10, 20, 45, and 47. However, we have noticed that only the finite difference reconstructed-based WENO scheme has been used to study the supersonic TGV problem so far, and interpolation-based counterparts have not been seen yet. In fact, compared to the reconstruction-based WENO scheme, the interpolation-based version can be paired with any approximate Riemann solver, thus having smaller numerical dissipation. Meanwhile, according to the report in Refs. 15, 21, and 22, it has better free-stream preserving features. Therefore, in order to fill this gap, this paper evaluates the performance of the interpolation-based WENO scheme for supersonic TGV problems (Mach numbers up to 2.5) based on the recently popular US-WENO scheme. In addition, in order to reduce the additional overhead caused by the interpolation-based US-WENO scheme, we develop an extremum property (EP)-based discontinuous sensor and hybrid scheme. This EP-based sensor does not contain artificially adjustable parameters and has good robustness for supersonic TGV flows, while significantly improving the computational efficiency of the developed interpolation-based US-WENO scheme.

This paper is organized as follows. In Sec. II, the governing equations and numerical methods are introduced. Then, a description and analysis of the results are given in Sec. III. Finally, concluding remarks are drawn in Sec. IV.

II. GOVERNING EQUATIONS AND NUMERICAL METHOD

A. Governing equations

In this paper, the compressible Euler and Navier–Stokes equations with the following conservative form are considered:

$$\frac{\partial \mathbf{U}(\mathbf{x},t)}{\partial t} + \nabla \cdot \left(\mathbf{F_c}(\mathbf{U}) - \frac{\sigma}{Re}\mathbf{F_v}(\mathbf{U},\nabla\mathbf{U})\right) = 0, \tag{1}$$

where $\mathbf{x} = (x, y, z)$ is the Cartesian coordinate, $\mathbf{U} = (\rho, \rho u, \rho v, \rho w, E)^{\mathrm{T}}$ is the vector of the conserved variables, ρ, $(u, v, w)^{\mathrm{T}}$, and p are the density, velocity vector, and pressure, respectively. In Eq. (1), $\sigma = 0$ represents the Euler equations, and $\sigma = 1$ represents the Navier–Stokes equations. Re is the Reynolds number. $E = \rho e + \frac{1}{2}\rho(u^2 + v^2 + w^2)$ is the total energy, where e is the specific internal energy. The equation of state is $p = (\gamma - 1)\rho e$, where $\gamma = 1.4$ is the ratio of the specific heats. $\mathbf{F_c}(\mathbf{U}) = (F_c^x, F_c^y, F_c^z)$ and $\mathbf{F_v}(\mathbf{U},\nabla\mathbf{U}) = (F_v^x, F_v^y, F_v^z)$ are the inviscid and viscous fluxes, and their specific expressions are

$$\begin{aligned}
F_c^x &= (\rho u, \rho u^2 + p, \rho uv, \rho uw, u(E+p))^{\mathrm{T}},\\
F_c^y &= (\rho v, \rho uv, \rho v^2 + p, \rho vw, v(E+p))^{\mathrm{T}},\\
F_c^z &= (\rho w, \rho uw, \rho vw, \rho w^2 + p, w(E+p))^{\mathrm{T}},\\
F_v^x &= (0, \tau_{xx}, \tau_{xy}, \tau_{xz}, u\tau_{xx} + v\tau_{xy} + w\tau_{xz} + q_x)^{\mathrm{T}},\\
F_v^y &= (0, \tau_{xy}, \tau_{yy}, \tau_{yz}, u\tau_{xy} + v\tau_{yy} + w\tau_{yz} + q_y)^{\mathrm{T}},\\
F_v^z &= (0, \tau_{xz}, \tau_{yz}, \tau_{zz}, u\tau_{xz} + v\tau_{yz} + w\tau_{zz} + q_z)^{\mathrm{T}}.
\end{aligned} \tag{2}$$

The viscous stress tensor τ_{ij} and the heat flux vector q_j in Eq. (2) are

$$\tau_{ij} = \mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) - \frac{2}{3}\mu\frac{\partial u_k}{\partial x_k}\delta_{ij}, \quad q_j = \frac{\mu}{(\gamma-1)Pr}\frac{\partial T}{\partial x_j}, \tag{3}$$

where the subscripts i, j, k refer to the Cartesian coordinate component $\mathbf{x} = (x, y, z)$, δ_{ij} is the Kronecker delta, (u_i, u_j, u_k) denotes the velocity components (u, v, w). T is the temperature, Pr is the Prandtl number, and μ represents the molecular viscosity computed by Sutherland's law.

This paper mainly considers the influence of the discretization method for convective terms on the results, while the viscous terms adopt a simple central difference scheme with the corresponding accuracy see Refs. 35 and 37 for more details. Therefore, the viscous terms in Eq. (1) will be temporarily ignored from now on. Meanwhile, for the convenience of introducing finite difference high-order WENO schemes, the one-dimensional scalar form of Eq. (1) will be considered,

$$U_t + f_x(U) = 0. \tag{4}$$

The calculation domain is divided into several intervals $I_i = [x_{i-1/2}, x_{i+1/2}]$, the half-node is located at $x_{i+1/2} = \frac{1}{2}(x_i + x_{i+1})$, and the space step is h. Then, the semi-discrete finite difference scheme of (4) at each grid point x_i can be reformulated as

$$\frac{\partial U}{\partial t} = -\frac{1}{h}(\hat{f}_{i+1/2} - \hat{f}_{i-1/2}), \tag{5}$$

where $\hat{f}_{i\pm 1/2}$ are numerical fluxes at the grid boundary $x_{i\pm 1/2}$, respectively.

The procedures of evaluating $\hat{f}_{i-1/2}$ and $\hat{f}_{i+1/2}$ are basically the same, so only the specific process of $\hat{f}_{i+1/2}$ will be introduced later. Furthermore, the discretization of convective terms in the y and z directions can also be obtained through the same steps, which are not covered here to save space. Generally, $\hat{f}_{i+1/2}$ in Eq. (5) can be obtained in two ways, namely, the reconstruction-based WENO schemes and the interpolation-based WENO schemes. However, Lusher and Sandham only considered and compared the performance of reconstruction-based WENO schemes for the supersonic TGV problem in Ref. 19. Therefore, this paper will evaluate in detail the impact of these two discrete methods on the compressible TGV problem. Below, we will briefly introduce these two different finite difference WENO schemes using the recently developed US-WENO scheme[48] as an example.

B. Reconstruction-based unequal-sized WENO scheme

In 2016, Zhu and Qiu[48] first proposed a fifth-order finite difference reconstruction-based WENO scheme based on unequal-sized stencils. This scheme is designed based on a five-point stencil and two two-point stencils, rather than using three equal-sized three-point stencils like the classical WENO-JS scheme. Compared to the classical WENO-JS scheme and its variants, the linear weights of the US-WENO scheme cannot be unique, making it easier to generalize to unstructured grids[50] and having better convergence for steady-state problems.[51,53,54] The process is briefly described below.

Step 1. For the stability of the scheme, the reconstruction-based WENO scheme generally performs flux splitting first. In this paper, the well-known Lax–Friedrichs flux splitting method is used,

$$f^{\pm}(U_i) = \frac{1}{2}[f(U_i) \pm \alpha U_i], \tag{6}$$

where α is set as $\max_U |f'(U)|$. Generally, the value of α can be based on the local reconstruction stencil or the whole range of U. However, the reconstruction-based WENO scheme generally chooses the latter to evaluate it for stability.

Next, we will be able to obtain $\hat{f}^{\pm}_{i+1/2}$ through $f^{\pm}(U_i)$. For convenience, $f^{\pm}(U_i)$ will be abbreviated as $f_i^{\pm}$ later, and only the process of obtaining $\hat{f}^{+}_{i+1/2}$ through $f^{+}(U_i)$ will be introduced. Similar procedures about $\hat{f}^{-}_{i+1/2}$ will not be repeated for saving space.

Step 2. A five-point stencil $S_1 = \{I_{i-2}, I_{i-1}, I_i, I_{i+1}, I_{i+2}\}$ and two two-point stencils $S_2 = \{I_{i-1}, I_i\}$, $S_3 = \{I_i, I_{i+1}\}$ are selected to obtain one quartic polynomial $p_1(x)$ and two linear polynomials $p_{2,3}(x)$, respectively. They satisfy the following constraints:

$$\begin{aligned}
&\frac{1}{h}\int_{x_j-h/2}^{x_j+h/2} p_1(x)dx = f_j^+, \quad j = i-2, \ldots, i+2, \\
&\frac{1}{h}\int_{x_j-h/2}^{x_j+h/2} p_2(x)dx = f_j^+, \quad j = i-1, i, \\
&\frac{1}{h}\int_{x_j-h/2}^{x_j+h/2} p_3(x)dx = f_j^+, \quad j = i, i+1.
\end{aligned} \tag{7}$$

Their specific expressions are

$$\begin{aligned}
p_1(x) = &\frac{9f_{i-2}^+ - 116f_{i-1}^+ + 2134f_i^+ - 116f_{i+1}^+ + 9f_{i+2}^+}{1920} \\
&+ \frac{5f_{i-2}^+ - 34f_{i-1}^+ + 34f_{i+1}^+ - 5f_{i+2}^+}{48}\frac{x - x_i}{h} \\
&+ \frac{f_{i-2}^+ - 12f_{i-1}^+ + 22f_i^+ - 12f_{i+1}^+ + f_{i+2}^+}{16}\left(\frac{x - x_i}{h}\right)^2 \\
&+ \frac{2f_{i-1}^+ - f_{i-2}^+ - 2f_{i+1}^+ + f_{i+2}^+}{12}\left(\frac{x - x_i}{h}\right)^3 \\
&+ \frac{-4f_{i-1}^+ + f_{i-2}^+ + 6f_i^+ - 4f_{i+1}^+ + f_{i+2}^+}{24}\left(\frac{x - x_i}{h}\right)^4, \\
p_2(x) &= f_i^+ + (f_i^+ - f_{i-1}^+)\left(\frac{x - x_i}{h}\right), \\
p_3(x) &= f_i^+ + (f_{i+1}^+ - f_i^+)\left(\frac{x - x_i}{h}\right).
\end{aligned} \tag{8}$$

Step 3. Choose a set of positive numbers $\gamma_n (n = 1, 2, 3)$ as the linear weights, which satisfy $\gamma_1 + \gamma_2 + \gamma_3 = 1$.

Step 4. The smoothness indicators $\beta_n (n = 1, 2, 3)$ corresponding to reconstruction polynomials $p_n(x) (n = 1, 2, 3)$ can be calculated using the following equation:

$$\beta_n = \sum_{\alpha=1}^{r} \int_{I_i} h^{2\alpha-1}\left(\frac{d^\alpha p_n(x)}{dx^\alpha}\right)^2 dx, \quad n = 1, 2, 3, \tag{9}$$

where r is the degree of the reconstruction polynomial, i.e., $r = 4$ for $p_1(x)$, $r = 1$ for $p_2(x)$ and $p_3(x)$. Their specific expressions are

$$\begin{aligned}
\beta_1 = \Big[&104963(f_{i-1}^+)^2 + 6908(f_{i-2}^+)^2 + 231153(f_i^+)^2 \\
&-299076f_i^+ f_{i+1}^+ + 104963(f_{i+1}^+)^2 + 67923f_i^+ f_{i+2}^+ \\
&-51001f_{i+1}^+ f_{i+2}^+ + 6908(f_{i+2}^+)^2 \\
&+f_{i-2}^+(67923f_i^+ - 38947f_{i+1}^+ + 8209f_{i+2}^+) \\
&-f_{i-1}^+(51001f_{i-2}^+ + 299076f_i^+ \\
&-179098f_{i+1}^+ + 38947f_{i+2}^+)\Big]/5040, \\
\beta_2 &= (f_i^+ - f_{i-1}^+)^2, \\
\beta_3 &= (f_{i+1}^+ - f_i^+)^2.
\end{aligned} \tag{10}$$

Step 5. Calculate the nonlinear weights,

$$\tau = \left(\frac{|\beta_1 - \beta_2| + |\beta_1 - \beta_3|}{2}\right)^2, \quad \overline{\omega}_n = \gamma_n\left(1 + \frac{\tau}{\varepsilon + \beta_n}\right),$$
$$\omega_n = \frac{\overline{\omega}_n}{\sum_{\ell=1}^{3}\overline{\omega}_\ell}, \quad n = 1, 2, 3, \tag{11}$$

where $\varepsilon = 10^{-6}$ is a small positive number chosen so that the denominator does not become zero.

Step 6. The final reconstruction-based US-WENO approximation for the numerical flux is

$$\hat{f}^{+}_{i+1/2} = \omega_1\left(\frac{1}{\gamma_1}p_1(x_{i+\frac{1}{2}}) - \frac{\gamma_2}{\gamma_1}p_2(x_{i+\frac{1}{2}}) - \frac{\gamma_3}{\gamma_1}p_3(x_{i+\frac{1}{2}})\right) + \omega_2 p_2(x_{i+\frac{1}{2}}) + \omega_3 p_3(x_{i+\frac{1}{2}}). \tag{12}$$

C. Interpolation-based unequal-sized WENO scheme

The reconstruction-based WENO scheme is intuitive, but there are a few available flux-splitting methods and its dissipation properties are not optimal. In contrast, the interpolation-based WENO scheme can be paired with any Riemann solver, resulting in smaller numerical dissipation. Therefore, many scholars have devoted their energy to it.[15,21,22] In Ref. 38, Wang *et al.* extended the US-WENO idea to this framework and developed an alternative US-WENO (AUS-WENO) scheme. Below is a brief review of its procedures.

Step 1. Following Refs. 15, 21, 22, and 38, the numerical flux $\hat{f}_{i+1/2}$ in Eq. (5) is first written in the following form:

$$\hat{f}_{i+1/2} = f_{i+1/2} - \frac{1}{24}h^2\left(\frac{\partial^2 f}{\partial x^2}\right)\bigg|_{x=x_{i+1/2}} + \frac{7}{5760}h^4\left(\frac{\partial^4 f}{\partial x^4}\right)\bigg|_{x=x_{i+1/2}}, \tag{13}$$

where the second and fourth derivative terms are approximated by the following central difference:

$$\frac{\partial^2 f}{\partial x^2}\bigg|_{x=x_{i+1/2}} = \frac{-5f_{i-2} + 39f_{i-1} - 34f_i - 34f_{i+1} + 39f_{i+2} - 5f_{i+3}}{48h^2} + O(h^6), \tag{14}$$

$$\frac{\partial^4 f}{\partial x^4}\bigg|_{x=x_{i+1/2}} = \frac{f_{i-2} - 3f_{i-1} + 2f_i + 2f_{i+1} - 3f_{i+2} + f_{i+3}}{2h^4} + O(h^6). \tag{15}$$

Then, only $f_{i+1/2}$ remains in Eq. (13), which is defined as

$$f_{i+1/2} = h(u^L_{i+1/2}, u^R_{i+1/2}), \tag{16}$$

where $h(u^L_{i+1/2}, u^R_{i+1/2})$ is a monotonic flux, which can be obtained by an arbitrary suitable Riemann solver, such as the LF (Lax–Friedrichs), HLL (Harten–Lax–van Leer), or HLLC (Harten–Lax–van Leer–Contact). As reported in Ref. 38, the LF method has better stability but greater dissipation, while the HLLC method has the best dissipation performance but the weakest stability. The HLL Riemann solver can balance the two aspects well. Therefore, the HLL method is used in this paper. In Eq. (16), $u^{L,R}_{i+1/2}$ represent the left and right numerical approximations at the grid boundary $x = x_{i+1/2}$. Next, we will introduce how they are calculated. To save space, we only introduce the approximate process of $u^L_{i+1/2}$, and the value of $u^R_{i+1/2}$ can be obtained similarly.

Step 2. Like the reconstruction-based US-WENO scheme, a five-point stencil S_1 and two two-point stencils $S_{2,3}$ are selected to obtain a fourth-degree polynomial and two linear polynomials. However, unlike the reconstruction-based US-WENO scheme, the interpolation-based US-WENO scheme directly obtains polynomials based on point values through the Lagrange interpolation method, which satisfies the following conditions:

$$\begin{aligned} q_1(x_j) &= u_j, \quad j = i-2, \ldots, i+2, \\ q_2(x_j) &= u_j, \quad j = i-1, i, \\ q_3(x_j) &= u_j, \quad j = i, i+1. \end{aligned} \tag{17}$$

Their specific expressions are

$$\begin{aligned} q_1(x) = u_i &- \frac{u_{i-2} - 8u_{i-1} + 8u_{i+1} - u_{i+2}}{12}\left(\frac{x - x_i}{h}\right) \\ &+ \frac{-u_{i-2} + 16u_{i-1} - 30u_i + 16u_{i+1} - u_{i+2}}{24}\left(\frac{x - x_i}{h}\right)^2 \\ &+ \frac{-u_{i-2} + 2u_{i-1} - 2u_{i+1} + u_{i+2}}{12}\left(\frac{x - x_i}{h}\right)^3 \\ &+ \frac{u_{i-2} - 4u_{i-1} + 6u_i - 4u_{i+1} + u_{i+2}}{24}\left(\frac{x - x_i}{h}\right)^4, \\ q_2(x) &= u_i + (u_i - u_{i-1})\left(\frac{x - x_i}{h}\right), \\ q_3(x) &= u_i + (u_{i+1} - u_i)\left(\frac{x - x_i}{h}\right). \end{aligned} \tag{18}$$

Step 3. Like the previously mentioned reconstruction-based US-WENO scheme, choose a set of arbitrary positive numbers $\gamma_n (n = 1, 2, 3)$ that sum to one as the linear weights.

Step 4. Calculate the smoothness indicators $\beta_n (n = 1, 2, 3)$ corresponding to the interpolation polynomials $q_n(x)$ using Eq. (9). Their specific expressions are

$$\begin{aligned} \beta_1 = [&82364u_{i-2}^2 + 1228889u_{i-1}^2 + 2695779u_i^2 + 1228889u_{i+1}^2 \\ &+ 82364u_{i+2}^2 - 601771u_{i-2}u_{i-1} + 799977u_{i-2}u_i \\ &- 461113u_{i-2}u_{i+1} + 98179u_{i-2}u_{i+2} - 3495756u_{i-1}u_i \\ &+ 2100862u_{i-1}u_{i+1} - 461113u_{i-1}u_{i+2} + 799977u_iu_{i+2} \\ &- 601771u_{i+1}u_{i+2} - 3495756u_iu_{i+1}]/60480, \\ \beta_2 &= (u_i - u_{i-1})^2, \\ \beta_3 &= (u_{i+1} - u_i)^2. \end{aligned} \tag{19}$$

Step 5. Compute the nonlinear weights $\omega_n (n = 1, 2, 3)$ using Eq. (11) in the procedures of reconstruction-based US-WENO scheme.

Step 6. The final fifth-order interpolation-based US-WENO approximation of $u(x)$ at $x = x_{i+1/2}$ is

$$u^L_{i+1/2} = \omega_1\left(\frac{1}{\gamma_1}q_1(x_{i+\frac{1}{2}}) - \frac{\gamma_2}{\gamma_1}q_2(x_{i+\frac{1}{2}}) - \frac{\gamma_3}{\gamma_1}q_3(x_{i+\frac{1}{2}})\right) + \omega_2 q_2(x_{i+\frac{1}{2}}) + \omega_3 q_3(x_{i+\frac{1}{2}}). \quad (20)$$

D. Hybrid interpolation-based unequal-sized WENO scheme

Generally, both reconstruction-based and interpolation-based WENO schemes are usually implemented in the local characteristic directions to reduce spurious oscillations. However, this process is very time-consuming. At the same time, the nonlinear weights calculation process of WENO schemes will also bring some computational consumption. The hybrid WENO scheme is an effective means to solve the above problems, which uses a more efficient linear method in smooth regions and a WENO scheme with good shock-capturing features near discontinuities. Therefore, many scholars have devoted their energy to this topic.[8,16–18,25,28] In this paper, we develop an EP-based (Extremum Property-based) discontinuity sensor based on the extremum property of the highest-degree polynomial and then design the corresponding hybrid interpolation-based US-WENO scheme. The detailed process of this algorithm is as follows.

Step 1. Assuming that the interpolation polynomial obtained by Eq. (17) based on the largest five-point stencil is $q_1(x) = \sum_{\ell=0}^{4} a_\ell \varphi_\ell(x)$, we can obtain the following first derivative polynomial:

$$q_1'(x) = \frac{4a_4}{h}\varphi_3(x) + \frac{3a_3}{h}\varphi_2(x) + \frac{2a_2}{h}\varphi_1(x) + \frac{a_1}{h}\varphi_0(x), \quad (21)$$

where a_ℓ and $\varphi_\ell(x) = (\frac{x-x_i}{h})^\ell$ ($\ell = 0, \ldots, 4$) denote polynomial coefficients and basis functions, respectively. Their expressions can be found in Eq. (18).

Step 2. Distinguish between smooth grid points and "troubled cells" based on the extremum properties of the interpolation polynomial $q_1(x)$, that is, whether the derivative polynomial $q_1'(x)$ has zero roots in the reconstruction interval. This idea has been discussed in Refs. 38 and 49, but it needs to find the specific zero root of $q_1'(x)$ by an explicit expression. Therefore, this has certain limitations. For example, it is only applicable to low-degree polynomials of one variable, while it is difficult to find an explicit expression for the roots of other forms of polynomials. In fact, we can determine whether the derivative polynomial contains zero roots based on whether the maximum and minimum values near the target grid point are different signs, without the need for an explicit expression. This idea has been well implemented in the hybrid reconstruction-based US-WENO schemes on structured[39] and unstructured[40] grids. Here, we extend this idea to the interpolation-based US-WENO scheme presented in this paper. Furthermore, we will only consider polynomial $\tilde{q}_1'(x) = 4a_4\varphi_3(x) + 3a_3\varphi_2(x) + 2a_2\varphi_1(x) + a_1\varphi_0(x)$ later, as its maximum and minimum values have the same sign as $q_1'(x)$.

Step 3. Evaluate the maximum and minimum values of $\tilde{q}_1'(x)$ within the interval $[x_{i-3/2}, x_{i+3/2}]$. We first notice the fact that $\tilde{q}_1'(x)$ is uniformly monotonic within the intervals $[x_{i-3/2}, x_i]$ and $[x_i, x_{i+3/2}]$, respectively. Therefore, we can estimate the maximum and minimum values based on this property. For the interval $[x_{i-3/2}, x_i]$,

$$\begin{aligned} MAX1 &= max\big(4a_4\varphi_3(x_{i-3/2}), 4a_4\varphi_3(x_i)\big) \\ &\quad + max\big(3a_3\varphi_2(x_{i-3/2}), 3a_3\varphi_2(x_i)\big) \\ &\quad + max\big(2a_2\varphi_1(x_{i-3/2}), 2a_2\varphi_1(x_i)\big) \\ &\quad + max\big(a_1\varphi_0(x_{i-3/2}), a_1\varphi_0(x_i)\big), \\ MIN1 &= min\big(4a_4\varphi_3(x_{i-3/2}), 4a_4\varphi_3(x_i)\big) \\ &\quad + min\big(3a_3\varphi_2(x_{i-3/2}), 3a_3\varphi_2(x_i)\big) \\ &\quad + min\big(2a_2\varphi_1(x_{i-3/2}), 2a_2\varphi_1(x_i)\big) \\ &\quad + min\big(a_1\varphi_0(x_{i-3/2}), a_1\varphi_0(x_i)\big), \end{aligned} \quad (22)$$

and for the interval $[x_i, x_{i+3/2}]$,

TABLE I. 3 D Euler equations. L^1 and L^∞ errors. $T = 2$.

Grid points	WENO-JS L^1 error	Order	L^∞ error	Order	US-WENO L^1 error	Order	L^∞ error	Order
20^3	2.63×10^{-4}		4.88×10^{-4}		3.67×10^{-5}		6.00×10^{-5}	
40^3	8.23×10^{-6}	5.00	1.60×10^{-5}	4.93	1.18×10^{-6}	4.96	1.89×10^{-6}	4.99
60^3	1.08×10^{-6}	5.01	2.07×10^{-6}	5.03	1.56×10^{-7}	4.99	2.50×10^{-7}	4.99
80^3	2.55×10^{-7}	5.02	4.82×10^{-7}	5.07	3.70×10^{-8}	4.99	5.94×10^{-8}	4.99
100^3	8.30×10^{-8}	5.03	1.54×10^{-7}	5.12	1.21×10^{-8}	5.00	1.95×10^{-8}	5.00
Grid points	AUS-WENO L^1 error	Order	L^∞ error	Order	HAUS-WENO L^1 error	Order	L^∞ error	Order
20^3	3.17×10^{-5}		6.18×10^{-5}		3.19×10^{-5}		5.46×10^{-5}	
40^3	1.01×10^{-6}	4.97	1.82×10^{-6}	5.09	1.01×10^{-6}	4.97	1.77×10^{-6}	4.99
60^3	1.34×10^{-7}	4.99	2.36×10^{-7}	5.03	1.34×10^{-7}	4.99	2.34×10^{-7}	4.99
80^3	3.18×10^{-8}	5.00	5.58×10^{-8}	5.02	3.18×10^{-8}	5.00	5.56×10^{-8}	4.99
100^3	1.04×10^{-8}	5.00	1.83×10^{-8}	5.00	1.04×10^{-8}	5.00	1.82×10^{-8}	5.00

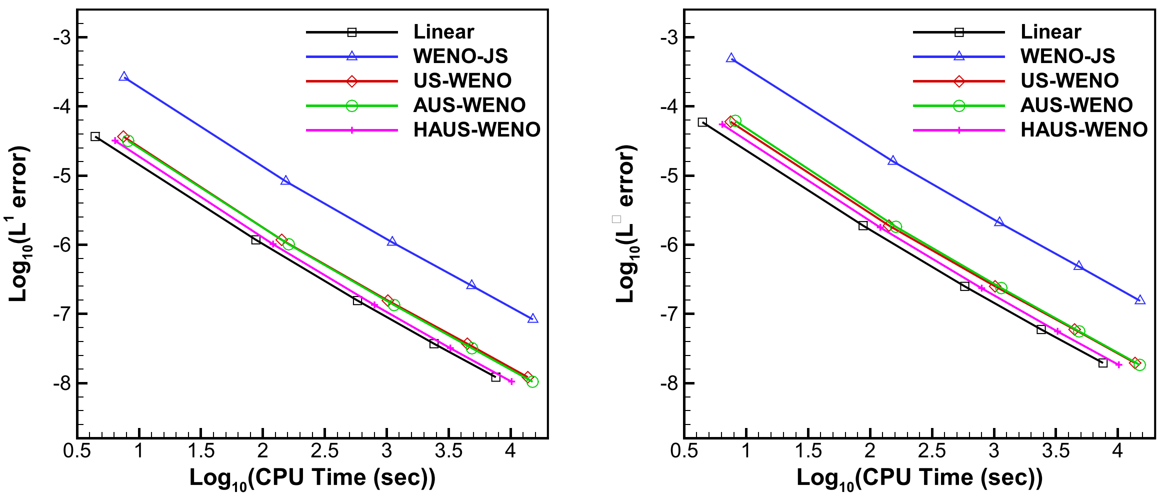


FIG. 1. A comparison of the CPU cost vs L^1 and L^∞ errors. Left: L^1 errors. Right: L^∞ errors. 3 D Euler equations.

$$
\begin{aligned}
MAX2 = {} & max\big(4a_4\varphi_3(x_i), 4a_4\varphi_3(x_{i+3/2})\big) \\
& + max\big(3a_3\varphi_2(x_i), 3a_3\varphi_2(x_{i+3/2})\big) \\
& + max\big(2a_2\varphi_1(x_i), 2a_2\varphi_1(x_{i+3/2})\big) \\
& + max\big(a_1\varphi_0(x_i), a_1\varphi_0(x_{i+3/2})\big), \\
MIN2 = {} & min\big(4a_4\varphi_3(x_i), 4a_4\varphi_3(x_{i+3/2})\big) \\
& + min\big(3a_3\varphi_2(x_i), 3a_3\varphi_2(x_{i+3/2})\big) \\
& + min\big(2a_2\varphi_1(x_i), 2a_2\varphi_1(x_{i+3/2})\big) \\
& + min\big(a_1\varphi_0(x_i), a_1\varphi_0(x_{i+3/2})\big).
\end{aligned} \tag{23}
$$

Step 4. If $(MAX1 \pm h)\cdot(MIN1 \pm h) > 0$ and $(MAX2 \pm h)\cdot(MIN2 \pm h) > 0$, then the target grid point is smooth, and a cheaper linear method is used, that is, $u^L_{i+1/2} = q_1(x_{i+\frac{1}{2}})$. Otherwise, if there are discontinuities near the target point, a nonlinear shock-capturing scheme is used, i.e., Eq. (20).

Remark. For the supersonic Taylor–Green vortex problem considered in this paper, the density ρ is used in steps 1–4 here to identify the troubled cells.

III. NUMERICAL TESTS

This section will analyze the performance of the interpolation-based WENO schemes for supersonic Taylor–Green vortex problems in detail. At the same time, the results of the reconstruction-based classical WENO-JS[14] and US-WENO[48] schemes are also listed for comparison. Equation (5) is advanced in time by a third-order Total-Variation-Diminishing (TVD) Runge–Kutta method[14,31] and the Courant–Friedrichs–Lewy (CFL) number of all WENO-type schemes considered in this paper is defined as 0.6. Specifically, the linear weights of all unequal-sized WENO schemes in this paper are defined as $(\gamma_1, \gamma_2, \gamma_3) = (0.98, 0.01, 0.01)$. For convenience, from now on, the interpolation-based alternative US-WENO scheme will be abbreviated

TABLE II. 3 D Navier–Stokes equations with $Re = 1000$. L^1 and L^∞ errors. $T = 2$.

	WENO-JS				US-WENO			
Grid points	L^1 error	Order	L^∞ error	Order	L^1 error	Order	L^∞ error	Order
10^3	8.78×10^{-2}		1.39×10^{-1}		8.22×10^{-2}		1.28×10^{-1}	
20^3	1.40×10^{-2}	2.65	2.01×10^{-2}	2.79	3.42×10^{-3}	4.59	5.55×10^{-3}	4.53
40^3	7.82×10^{-4}	4.17	1.18×10^{-3}	4.09	1.14×10^{-4}	4.91	1.85×10^{-4}	4.90
80^3	2.55×10^{-5}	4.94	4.48×10^{-5}	4.71	3.63×10^{-6}	4.97	5.84×10^{-6}	4.99
	AUS-WENO				HAUS-WENO			
Grid points	L^1 error	Order	L^∞ error	Order	L^1 error	Order	L^∞ error	Order
10^3	8.94×10^{-2}		1.39×10^{-1}		5.26×10^{-2}		8.13×10^{-2}	
20^3	3.60×10^{-3}	4.64	6.31×10^{-3}	4.46	2.83×10^{-3}	4.22	4.56×10^{-3}	4.16
40^3	9.64×10^{-5}	5.22	1.84×10^{-4}	5.10	9.69×10^{-5}	4.87	1.68×10^{-4}	4.76
80^3	3.08×10^{-6}	4.97	5.47×10^{-6}	5.07	3.08×10^{-6}	4.98	5.37×10^{-6}	4.96

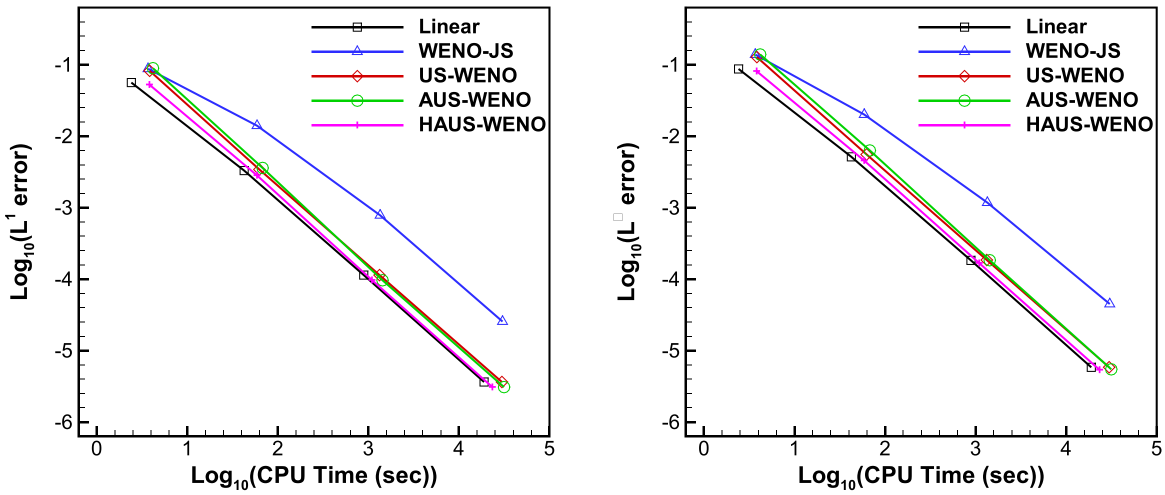


FIG. 2. A comparison of the CPU cost vs L^1 and L^∞ errors. Left: L^1 errors. Right: L^∞ errors. 3 D Navier–Stokes equations.

as AUS-WENO, and the hybrid AUS-WENO scheme will be abbreviated as HAUS-WENO.

A. Accuracy test for three-dimensional inviscid and viscous smooth flows

To validate the numerical accuracy of the proposed WENO schemes, the three-dimensional Eq. (1) is considered here. The computational domain is $[0, 2\pi] \times [0, 2\pi] \times [0, 2\pi]$ and the initial conditions are as follows:

$$\begin{aligned} &\rho(x, y, z, 0) = 1 + 0.2\sin(x + y + z), \quad u(x, y, z, 0) = 0.3, \\ &v(x, y, z, 0) = 0.3, \quad w(x, y, z, 0) = 0.4, \quad p(x, y, z, 0) = 1. \end{aligned} \tag{24}$$

There are periodic boundaries all around. The final time is $t = 2$ and the time step $\Delta t \approx O(h^{5/3})$, which ensures that the spatial errors dominate in this accuracy test.

For the inviscid case, the exact solutions of density ρ is $\rho(x, y, z, t) = 1 + 0.2\sin(x + y + z - t)$. In Table I, the L^1 errors, L^∞ errors, and corresponding convergence orders for different WENO-type schemes are given, respectively. It can be seen that the US-WENO scheme has smaller errors compared to the classical WENO-JS scheme at the same mesh level, although they both achieve the expected fifth-order accuracy. Meanwhile, owing to the use of a less dissipative Riemann approximate solver, the errors of the interpolation-based US-WENO schemes are slightly smaller than those of the US-WENO scheme under the same conditions. The same conclusion can also be drawn from Fig. 1, which shows a comparison of the central processing

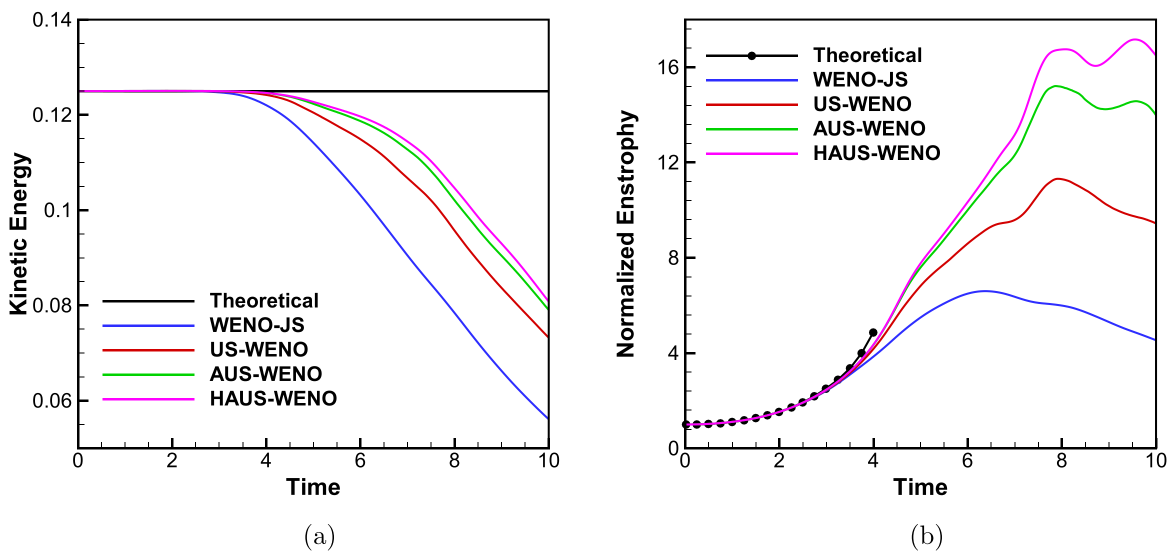


FIG. 3. Comparison of kinetic energy and enstrophy vs time curves for three-dimensional inviscid TGV problem with $Ma_{ref} = 0.1$. (a) Kinetic energy and (b) enstrophy. 128 × 128 × 128 grid points.

unit (CPU) time vs L^1 and L^∞ errors for different WENO-type schemes. In terms of computational efficiency, the US-WENO and WENO-JS schemes have similar computation times, while the AUS-WENO scheme is slightly slower due to the additional steps generated in its procedures. Relatively speaking, the HAUS-WENO scheme has higher computational efficiency due to its avoidance of tedious local characteristic decompositions and nonlinear weights calculation processes.

Subsequently, we evaluate the numerical accuracy of our method for three-dimensional Navier–Stokes equations. The initial and boundary conditions are the same as in the inviscid case, and the remaining parameters in Eq. (1) are $\mu = 0.0005$, $Pr = 2/3$, $\gamma = 1.4$, $Re = 1000$. In particular, due to viscous effects, the exact solution for the inviscid case is no longer satisfied, and therefore the asymptotic convergence errors[35,37,46] are used here to evaluate the convergence orders. Again, the L^1 errors, L^∞ errors, and corresponding convergence orders for different WENO-type schemes are shown in Table II. In Fig. 2, the CPU time vs L^1 and L^∞ errors of different WENO-type schemes for the Navier–Stokes equations were compared. From these tables and figures, we can draw the same conclusion as in the inviscid case, which will not be repeated here to save space.

B. Taylor–Green vortex flow setup and inviscid case verification

The TGV problem is a well-known test case to assess the dissipation performance of high-order computational fluid dynamics (CFD)

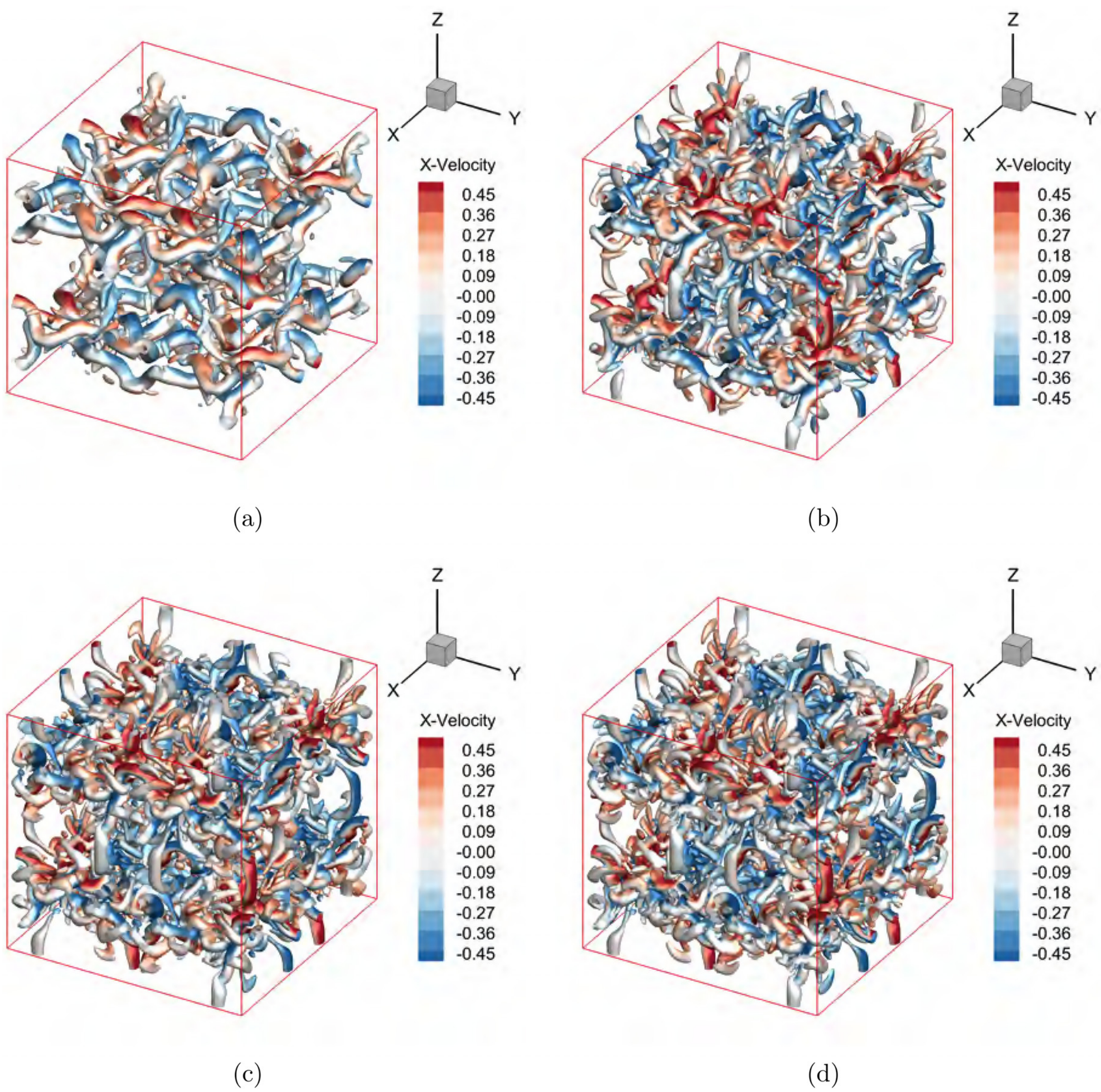


FIG. 4. *Q*-criterion ($Q = 1.0$ colored by the *x*-velocity) iso-surfaces of vortical structures obtained by different WENO-type schemes for inviscid TGV problem with $Ma_{ref} = 0.1$. (a) WENO-JS, (b) US-WENO, (c) AUS-WENO, and (d) HAUS-WENO. $128 \times 128 \times 128$ grid points.

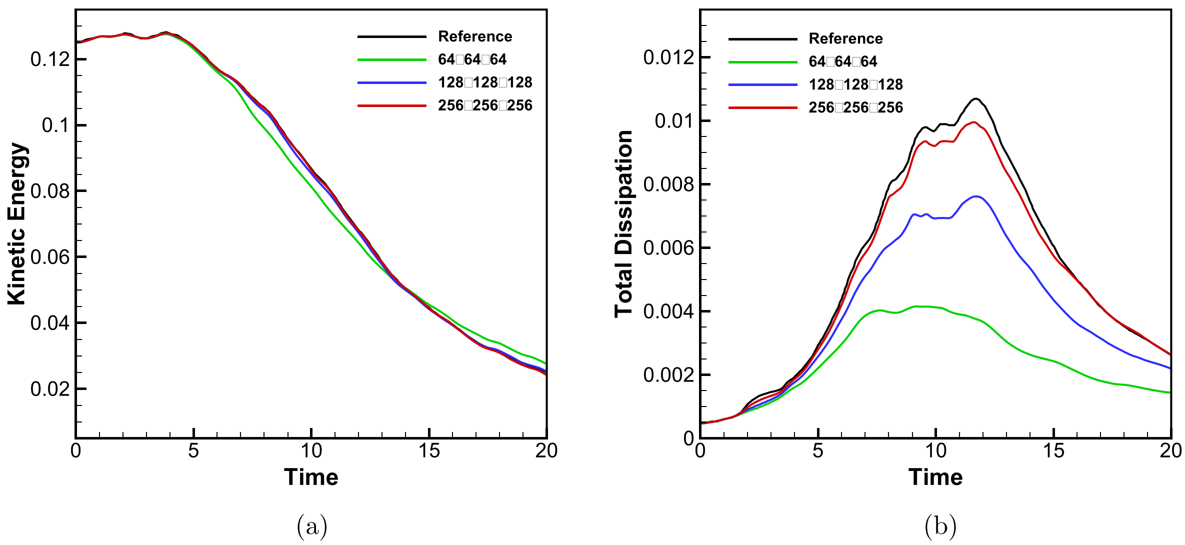


FIG. 5. Grid sensitivity of the compressible TGV problem. (a) Kinetic energy. (b) Total dissipation. HAUS-WENO scheme. $Ma_{ref} = 1.25$ and $Re = 1600$.

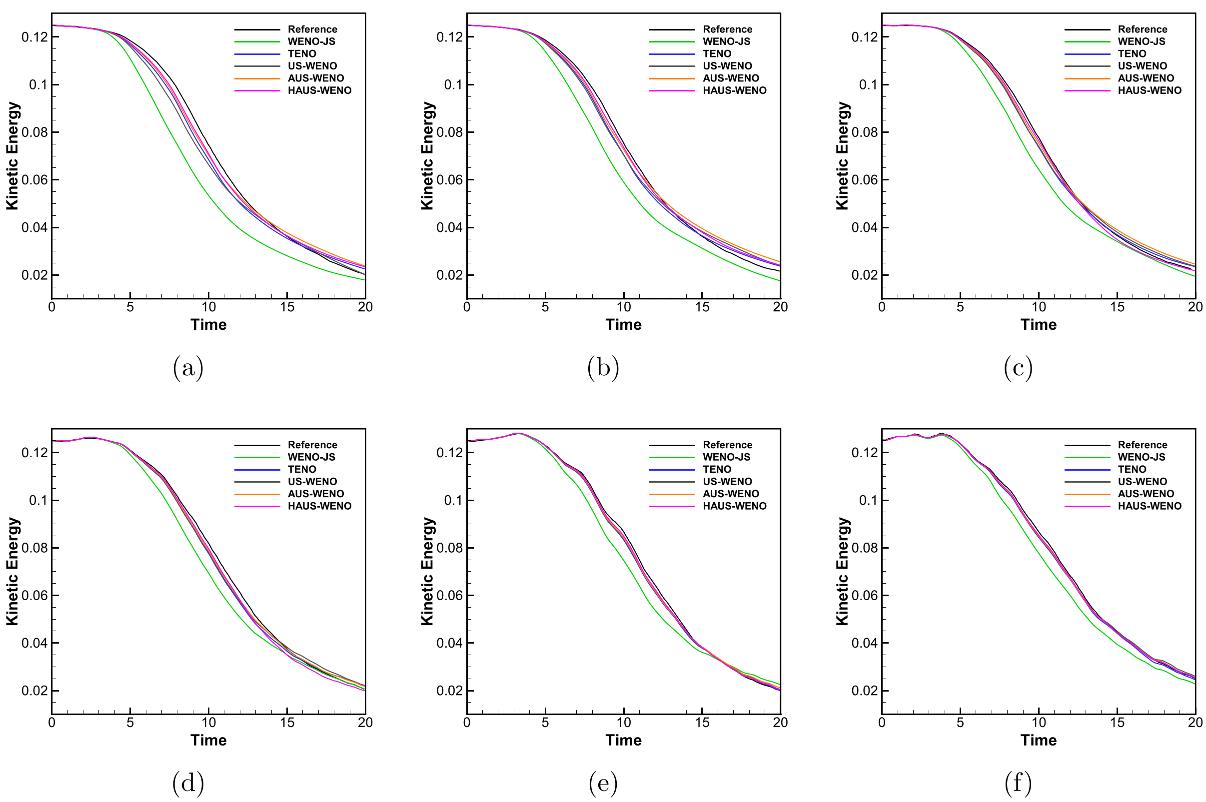


FIG. 6. Comparison of kinetic energy vs time curves for viscous TGV problem with different Mach numbers. (a) $Ma_{ref} = 0.1$, (b) $Ma_{ref} = 0.25$, (c) $Ma_{ref} = 0.5$, (d) $Ma_{ref} = 0.75$, (e) $Ma_{ref} = 1.0$, and (f) $Ma_{ref} = 1.25$. $Re = 1600$. $128 \times 128 \times 128$ grid points.

methods. The computational domain is $[0, 2\pi L] \times [0, 2\pi L] \times [0, 2\pi L]$ and the initial conditions are as follows:[19,45]

$$
\begin{aligned}
\rho(\mathbf{x}, t=0) &= 1 + \frac{C}{16}\left[\cos\left(\frac{2x}{L}\right) + \cos\left(\frac{2y}{L}\right)\right]\left[\cos\left(\frac{2z}{L}\right) + 2\right], \\
u(\mathbf{x}, t=0) &= V_0 \sin\left(\frac{x}{L}\right)\cos\left(\frac{y}{L}\right)\cos\left(\frac{z}{L}\right), \\
v(\mathbf{x}, t=0) &= -V_0 \cos\left(\frac{x}{L}\right)\sin\left(\frac{y}{L}\right)\cos\left(\frac{z}{L}\right), \\
w(\mathbf{x}, t=0) &= 0, \quad T(\mathbf{x}, t=0) = 1, \\
p(\mathbf{x}, t=0) &= \frac{\rho T}{\gamma Ma_{ref}^2},
\end{aligned} \tag{25}
$$

where $L = 1$, $V_0 = 1$, and Ma_{ref} is Mach number. The coefficient $C = \gamma Ma_{ref}^2$ for the inviscid case and $Re = 1600$, and $C = 1$ for $Re = 400$. The periodic boundary conditions are defined in all directions. To better quantify the effect of the numerical schemes on the TGV problem, the following volume-averaged quantities related to turbulent motion will be calculated:

- The first is the kinetic energy E_k, which represents large-scale turbulent motion,

$$
E_k = \frac{1}{\rho_{ref}|V|}\int_V \frac{1}{2}\rho u_j \cdot u_j dV. \tag{26}
$$

- The enstrophy ζ is given by

$$
\zeta = \frac{1}{\rho_{ref}|V|}\int_V \frac{1}{2}\rho \omega_j \cdot \omega_j dV. \tag{27}
$$

- The total viscous dissipation rate is given by

$$
\begin{aligned}
\varepsilon^T &= \varepsilon^S + \varepsilon^D \\
&= \frac{1}{\rho_{ref} Re}\int_V \mu\left(\varepsilon_{ijk}\frac{\partial u_k}{\partial u_j}\right)^2 dV + \frac{4}{3\rho_{ref} Re}\int_V \mu\left(\frac{\partial u_j}{\partial u_j}\right)^2 dV,
\end{aligned} \tag{28}
$$

where V is the control volume, $|V|$ is the volume of V, u_j is the velocity vector, and ω_j is the vorticity vector. In Eq. (28), ε^S and ε^D correspond to the solenoidal part and dilatational component of kinetic energy dissipation, which are directly related to the vortical motion and compressibility effects of the flow. The Q-criterion iso-surfaces of vortical structures are computed by

$$
Q = \frac{1}{2}\left(\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial y}\right)^2 + \left(\frac{\partial w}{\partial z}\right)^2\right) - \frac{\partial u}{\partial y}\frac{\partial v}{\partial x} - \frac{\partial u}{\partial z}\frac{\partial w}{\partial x} - \frac{\partial v}{\partial z}\frac{\partial w}{\partial y}. \tag{29}
$$

To verify the solver used in this paper and compare different schemes, we first consider the inviscid TGV problem with a Mach number $Ma_{ref} = 0.1$. For the inviscid case, Shu *et al.*[33] pointed out that

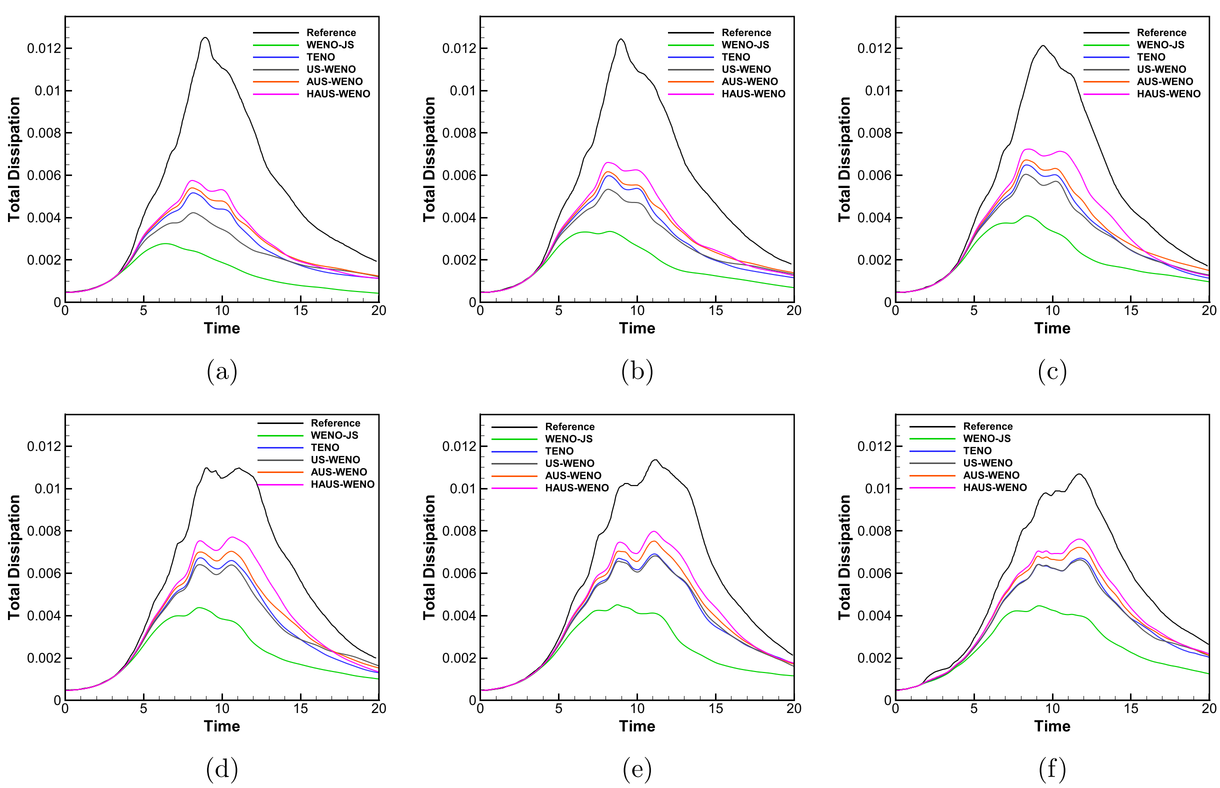
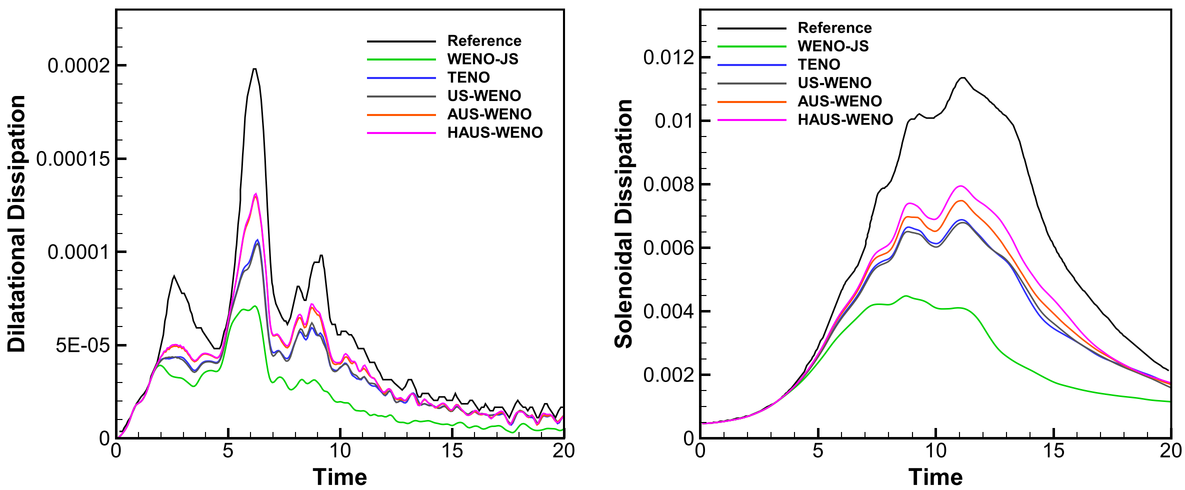


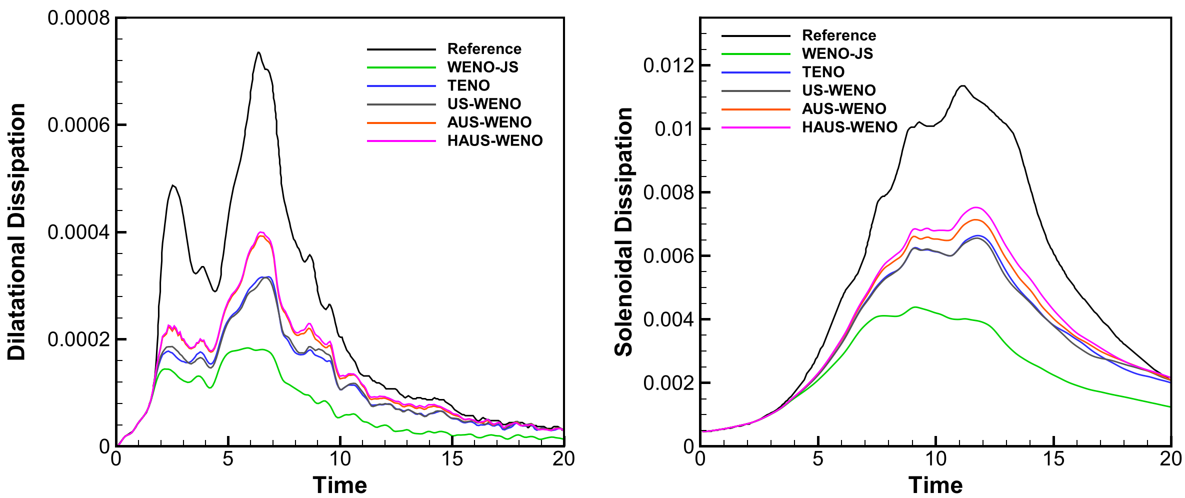
FIG. 7. Comparison of total viscous dissipation vs time curves for viscous TGV problem with different Mach numbers. (a) $Ma_{ref} = 0.1$, (b) $Ma_{ref} = 0.25$, (c) $Ma_{ref} = 0.5$, (d) $Ma_{ref} = 0.75$, (e) $Ma_{ref} = 1.0$, and (f) $Ma_{ref} = 1.25$. $Re = 1600$. $128 \times 128 \times 128$ grid points.



(a) (b)

FIG. 8. Comparison of dilatational and solenoidal dissipation vs time curves for different WENO-type schemes. (a) Dilatational component ε^D, (b) solenoidal part ε^S. $128 \times 128 \times 128$ grid points. $Ma_{ref} = 1.0$.

the theoretical value of kinetic energy does not change with time, that is, $E_k = 0.125$. Therefore, we can effectively evaluate the performance of different nonlinear schemes based on this property. Figure 3(a) shows the curves of kinetic energy E_k over time for all WENO-type schemes at the same grid level (i.e., $128 \times 128 \times 128$ grid points). It can be seen that the reconstruction-based WENO-JS scheme dissipates the fastest under the same conditions, while the US-WENO scheme using the same reconstruction procedures performs relatively better. Compared to the reconstruction-based US-WENO scheme, the interpolation-based US-WENO scheme is closer to the theoretical value, owing to the use of the HLL Riemann solver with lower dissipation. The HAUS-WENO scheme performs best among several considered WENO-type schemes because it uses a linear method with less dissipation in some areas. In Fig. 3(b), the mean enstrophy obtained by Eq. (27) for different WENO-type schemes is given, and the analytical solutions (before time $t \approx 4$) by Brachet *et al.* reported in Ref. 4 are also provided for comparison. The results show that the proposed method agrees well with the analytical solutions in the early stage, and then the enstrophy increases rapidly as the vortex begins to break up. In Fig. 4, the Q-criterion iso-surfaces of vortical structures obtained by different WENO-type schemes are shown at $t = 10$, which is colored by the u component velocity. We can clearly see that the



(a) (b)

FIG. 9. Comparison of dilatational and solenoidal dissipation vs time curves for different WENO-type schemes. (a) Dilatational component ε^D, (b) solenoidal part ε^S. $128 \times 128 \times 128$ grid points. $Ma_{ref} = 1.25$.

classical WENO-JS scheme dissipates many small structures, while other US-WENO schemes perform better at the same mesh level. Meanwhile, the two interpolation-based US-WENO schemes seem to be slightly better than the reconstruction-based US-WENO scheme. This also confirms the conclusion obtained in Fig. 3(a).

C. Viscous TGV problem with Mach numbers over the range of $M_{ref} = [0.1, 1.25]$

Here, we continue to consider the viscous TGV problem with the Mach numbers ranging from the standard subsonic case $Ma_{ref} = 0.1$ to the supersonic case $Ma_{ref} = 1.25$. In this case, the presence of shocklets and the compression effects will become an important factor affecting the development of turbulence. The Reynolds number is 1600, and the initial conditions are the same as in the inviscid case. The calculation process is advanced to a non-dimensional time $t = 20$.

First, considering the $Re = 1600$ and $Ma_{ref} = 1.25$ case, Fig. 5 shows the results of the HAUS-WENO scheme at different levels of grids, and the reference results of the sixth-order TENO scheme with $512 \times 512 \times 512$ grid points in Ref. 19 are also presented for comparison. It can be seen that the proposed method can gradually converge to the reference solutions. In order to evaluate the performance of

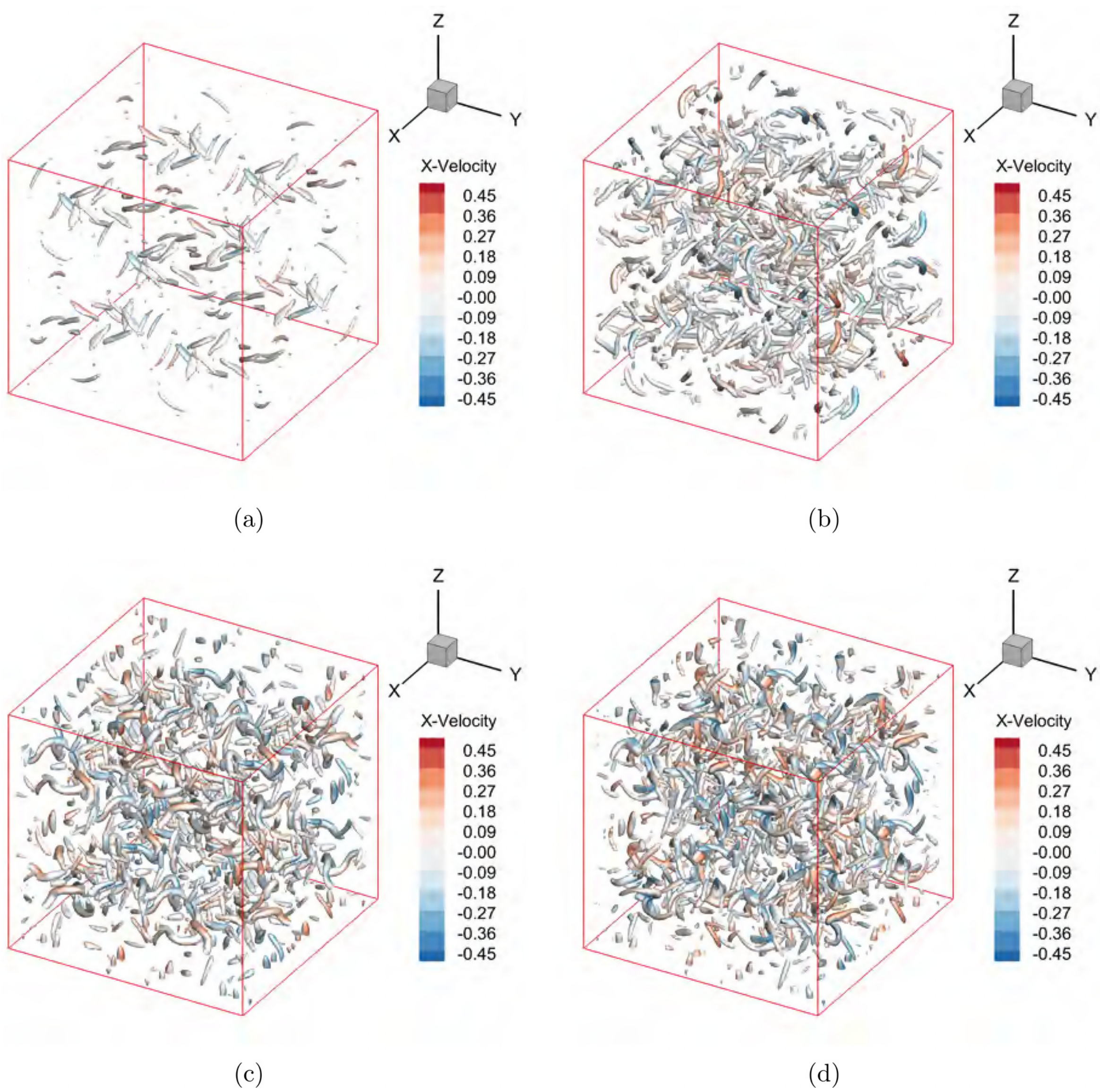


FIG. 10. *Q*-criterion ($Q = 2.5$ colored by the *x*-velocity) iso-surfaces of vortical structures obtained by different WENO-type schemes for viscous TGV problem with $Ma_{ref} = 1.25$. (a) WENO-JS, (b) US-WENO, (c) AUS-WENO, and (d) HAUS-WENO. $T = 20$. $Re = 1600$. $128 \times 128 \times 128$ grid points.

different schemes with limited grid resolution, the coarser $128 \times 128 \times 128$ grid points are mainly used later.

Figure 6 shows the curves of kinetic energy over time for $128 \times 128 \times 128$ grid points at different Mach numbers, and the reference solutions are derived from the results of $512 \times 512 \times 512$ grid points in Ref. 19. Meanwhile, the calculation results of the established fifth-order reconstruction-based TENO scheme under the same conditions are also provided in these figures as a reference. From these figures, it can be seen that the kinetic energy shows a monotonically decreasing trend at Mach numbers of 0.1, 0.25, and 0.5. However, as the Mach number increases, there is an area of increased kinetic energy rather than a continuous decline. This is consistent with the phenomenon observed in Refs. 7, 19, and 23. As the Mach number increases, the average pressure work may exceed the viscous dissipation rate, and the internal energy is converted to kinetic energy in the early stages. Therefore, the increase in kinetic energy is most obvious at $Ma_{ref} = 1.25$. Meanwhile, the WENO-JS scheme performs the worst for all the Mach numbers considered, indicating that it has the greatest numerical dissipation among all considered WENO-type schemes. At Mach numbers $Ma_{ref} = 0.1, 0.25, 0.5$, the results of the HAUS-WENO scheme are closer to the reference solutions due to their smaller numerical dissipation. However, for higher Mach numbers, all US-WENO schemes perform similarly, which means that the slight difference in numerical dissipation between them is not sufficient to distinguish performance.

In Fig. 7, the curves of total viscous dissipation [computed by Eq. (28)] over time at different Mach numbers are shown. Similarly, the reference solutions are derived from the sixth-order scheme under the $512 \times 512 \times 512$ grid points in Ref. 19. First, compared with the kinetic energy curves, the total dissipation rate curves can observe a greater difference for the same range of Mach numbers, and the HAUS-WENO scheme with lower dissipation performs the best among all WENO schemes considered here. At the same time, it can be observed that the peak of dissipation rate is delayed with increase in Mach number, and the peak value tends to decrease. At Mach numbers of 0.75, 1.0, and 1.25, a more pronounced double peak can be observed, which has been reported in Ref. 19. After $t = 12$, the total dissipation rate of Mach numbers 0.1, 0.25, and 0.5 rapidly decreases, while higher Mach number cases are relatively flat. Specifically, at $Ma_{ref} = 1.25$, shocklets are generated in the early stages due to compression effects, resulting in an additional peak during the $2 < t < 4$ time period, which is a phenomenon that does not exist at lower Mach numbers. However, it seems that none of the methods in this paper captured this phenomenon, which is caused by insufficient grid points, which is explained in detail in Ref. 19. Specifically, the dilatation and solenoidal parts of the total viscous dissipation for $Ma_{ref} = 1.0$ and

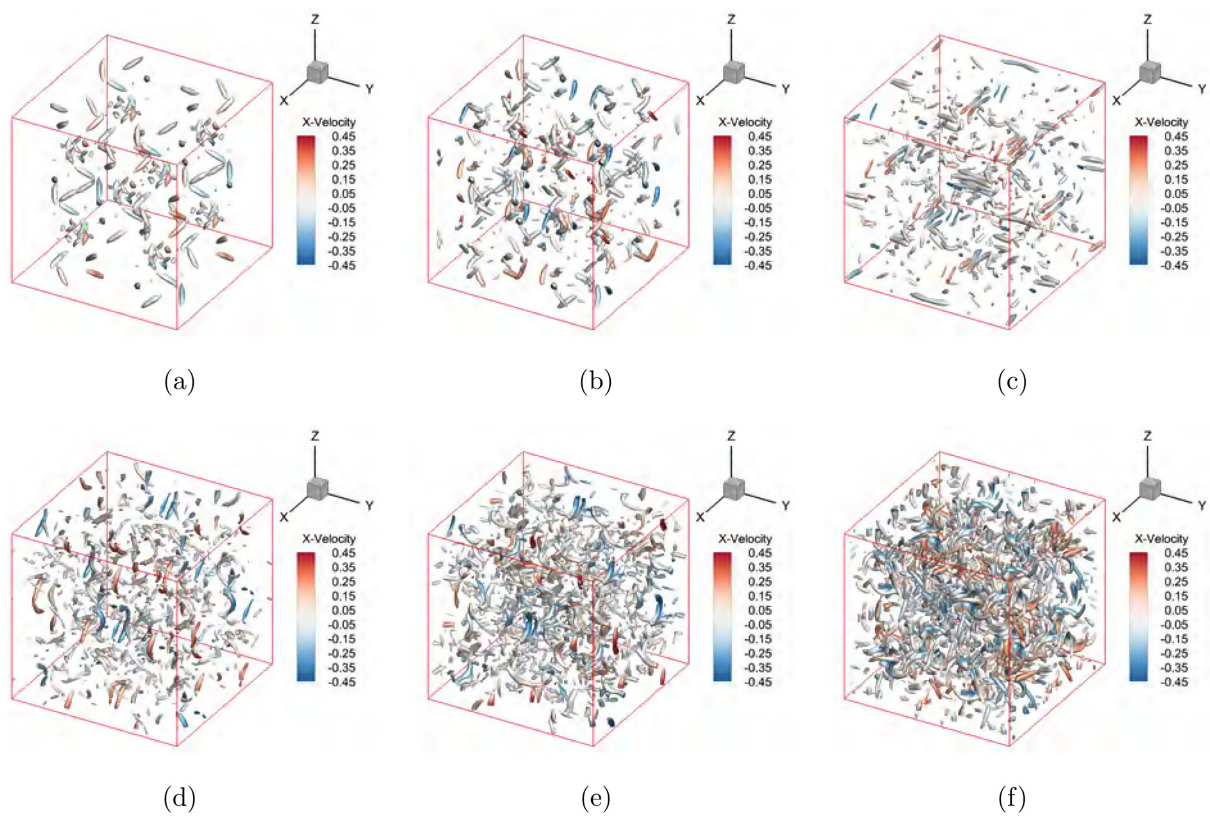


FIG. 11. *Q*-criterion ($Q = 2.0$ colored by the *x*-velocity) iso-surfaces of vortical structures obtained by HAUS-WENO scheme for viscous TGV problem with different Mach numbers. (a) $Ma_{ref} = 0.1$, (b) $Ma_{ref} = 0.25$, (c) $Ma_{ref} = 0.5$, (d) $Ma_{ref} = 0.75$, (e) $Ma_{ref} = 1.0$, and (f) $Ma_{ref} = 1.25$. $Re = 1600$. $128 \times 128 \times 128$ grid points.

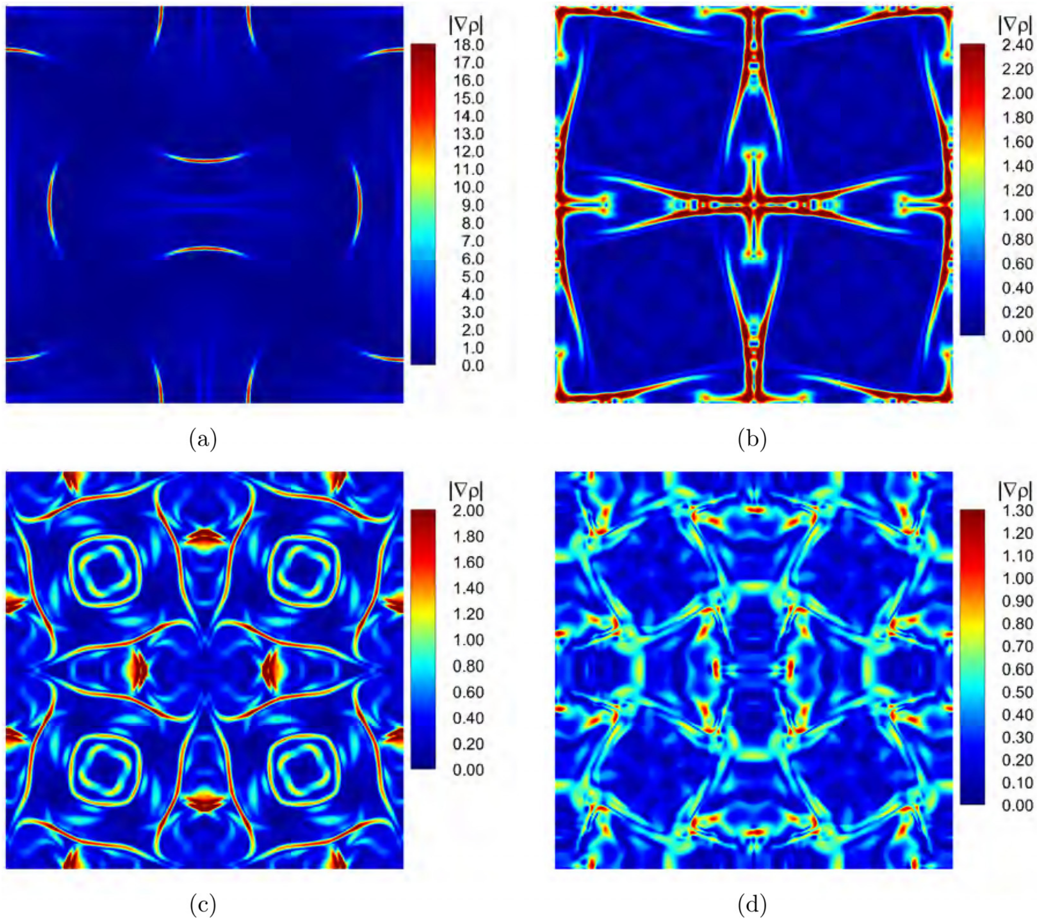


FIG. 12. Density gradients $|\nabla\rho|$ obtained by the HAUS-WENO scheme with 256^3 grid points at different times. (a) t = 2, (b) t = 7, (c) t = 15, and (d) t = 20. $Ma_{ref} = 1.25$ and $Re = 1600$.

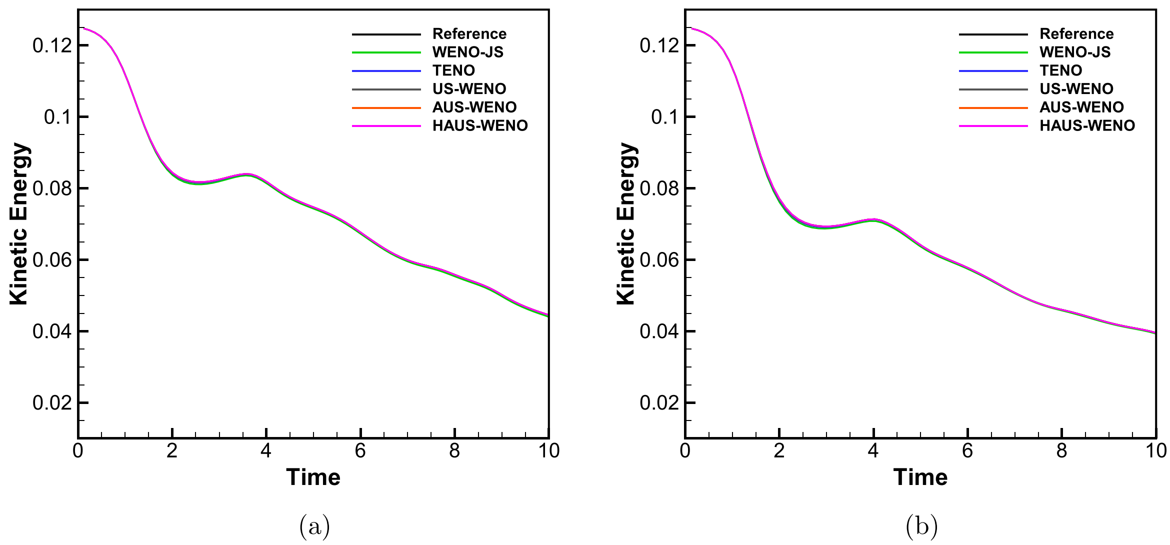


FIG. 13. Comparison of kinetic energy vs time curves for viscous TGV problem with different Mach numbers. (a) $Ma_{ref} = 2.0$, (b) $Ma_{ref} = 2.5$. $Re = 400$. 128 × 128 × 128 grid points.

$Ma_{ref} = 1.25$ are shown in Figs. 8 and 9, respectively. Compared to solenoidal dissipation, the magnitude of dilatational dissipation is very small, and two distinct peaks appeared in the early stages. This is because many shocklets are generated during the early stages at high Mach numbers. We also found that at the same mesh level, the HAUS-WENO scheme performs better and is closer to the reference solutions, especially for solenoidal dissipation.

To further compare the dissipation of different WENO schemes for the supersonic TGV problem, the Q-criterion iso-surfaces of vortical structures for $Ma_{ref} = 1.25$ are given in Fig. 10, which are colored by the u component velocity. It is obvious that the classical WENO-JS scheme has greater dissipation, followed by the reconstruction-based US-WENO scheme, while the interpolation-based AUS-WENO and HAUS-WENO schemes have smaller numerical dissipation and can capture more small structures at the same mesh level. Figure 11 shows the Q-criterion iso-surfaces obtained by the HAUS-WENO scheme for different Mach numbers at $t = 20$. It can be observed that the generation of shocklets makes small-scale turbulence more complex with increase in Mach number. This also means that as the Mach number increases, the coexistence and interaction of vortex dynamics, shocklets, and small-scale turbulence become more challenging for high-order numerical schemes.

In order to clarify the evolution of shock waves in the supersonic TGV problem ($Ma_{ref} = 1.25$), Fig. 12 shows the density gradient

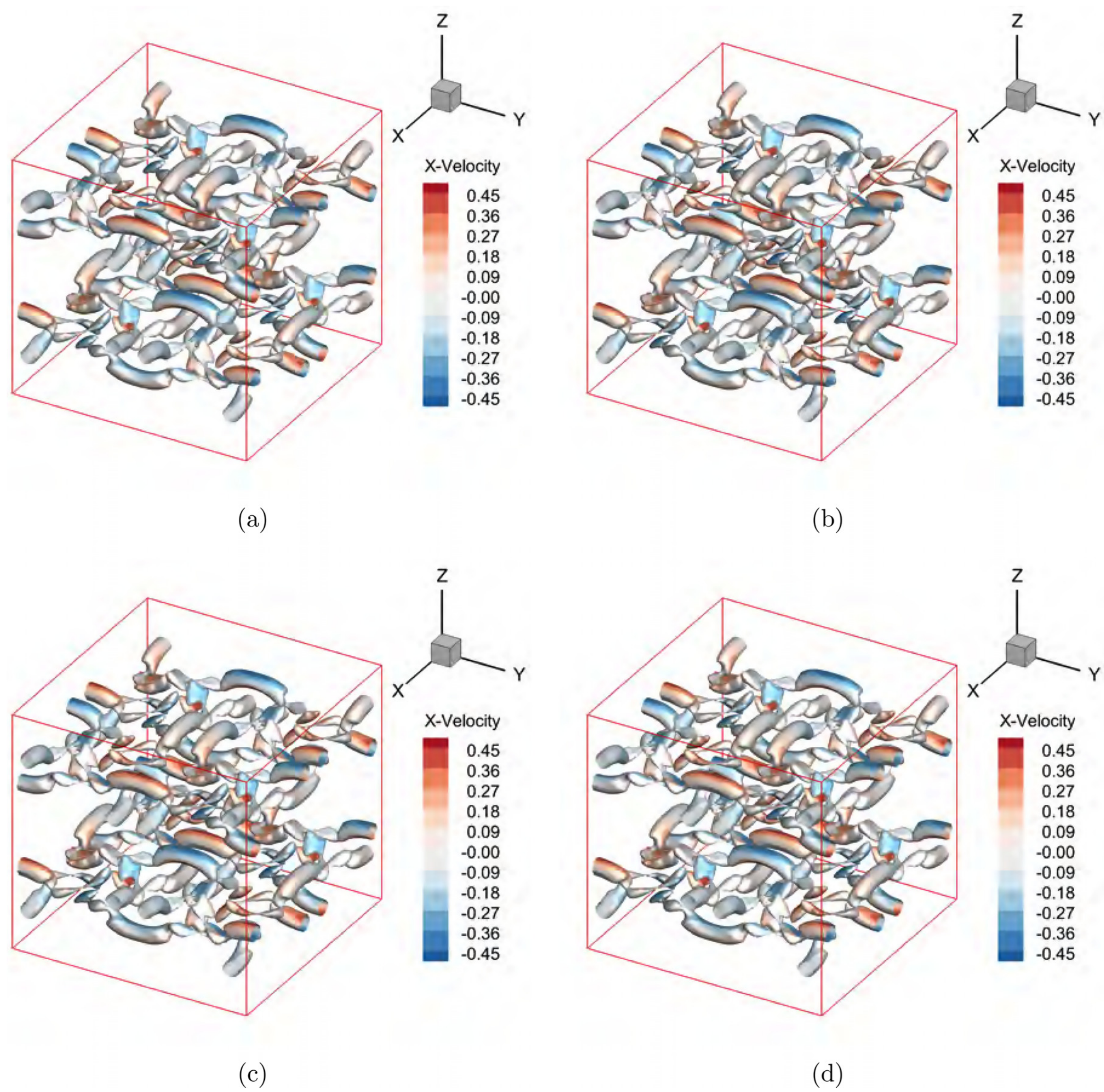


FIG. 14. Q-criterion ($Q = 1.0$ colored by the x-velocity) iso-surfaces of vortical structures obtained by different WENO-type schemes for viscous TGV problem with $Ma_{ref} = 2.0$. (a) WENO-JS, (b) US-WENO, (c) AUS-WENO, and (d) HAUS-WENO. $Re = 400$. $128 \times 128 \times 128$ grid points.

(calculated by $|\nabla\rho| = \sqrt{(\frac{\partial\rho}{\partial x})^2 + (\frac{\partial\rho}{\partial y})^2 + (\frac{\partial\rho}{\partial z})^2}$) of the HAUS-WENO scheme at simulation times $t = 2, 7, 15, 20$. The grid points are $256 \times 256 \times 256$. As shown in Fig. 12(a), due to the symmetry of the initial conditions, eight shock waves can be clearly observed at time $t = 2$. Due to the Mach number reaching 1.25, the density gradient's magnitude is even greater, which cannot be seen in subsonic TGV flows. At $t = 7$, as shown in Fig. 12(b), the shock waves form a slender structure after reflection, and the density gradient's magnitude begins to decrease. As shown in Fig. 12(c), the flow begins to become a subsonic state, but shocklets can still be seen propagating, while the relative motion of the fluid forms four counter-rotating flows. At the final moment, the large vortex completely breaks down and the flow field becomes relatively chaotic, as shown in Fig. 12(d). Although the three-dimensional perspective in Fig. 10(d) looks very chaotic, Fig. 12(d) shows that the results obtained by our method still maintain good symmetry, which is an important requirement for high-order numerical algorithms.

D. Viscous TGV problem with $M_{ref} = 2.0$ and $M_{ref} = 2.5$

The TGV problem with higher Mach numbers poses a more severe challenge to high-order numerical methods. Therefore, we will continue to consider the cases of Mach numbers 2.0 and 2.5 here. The

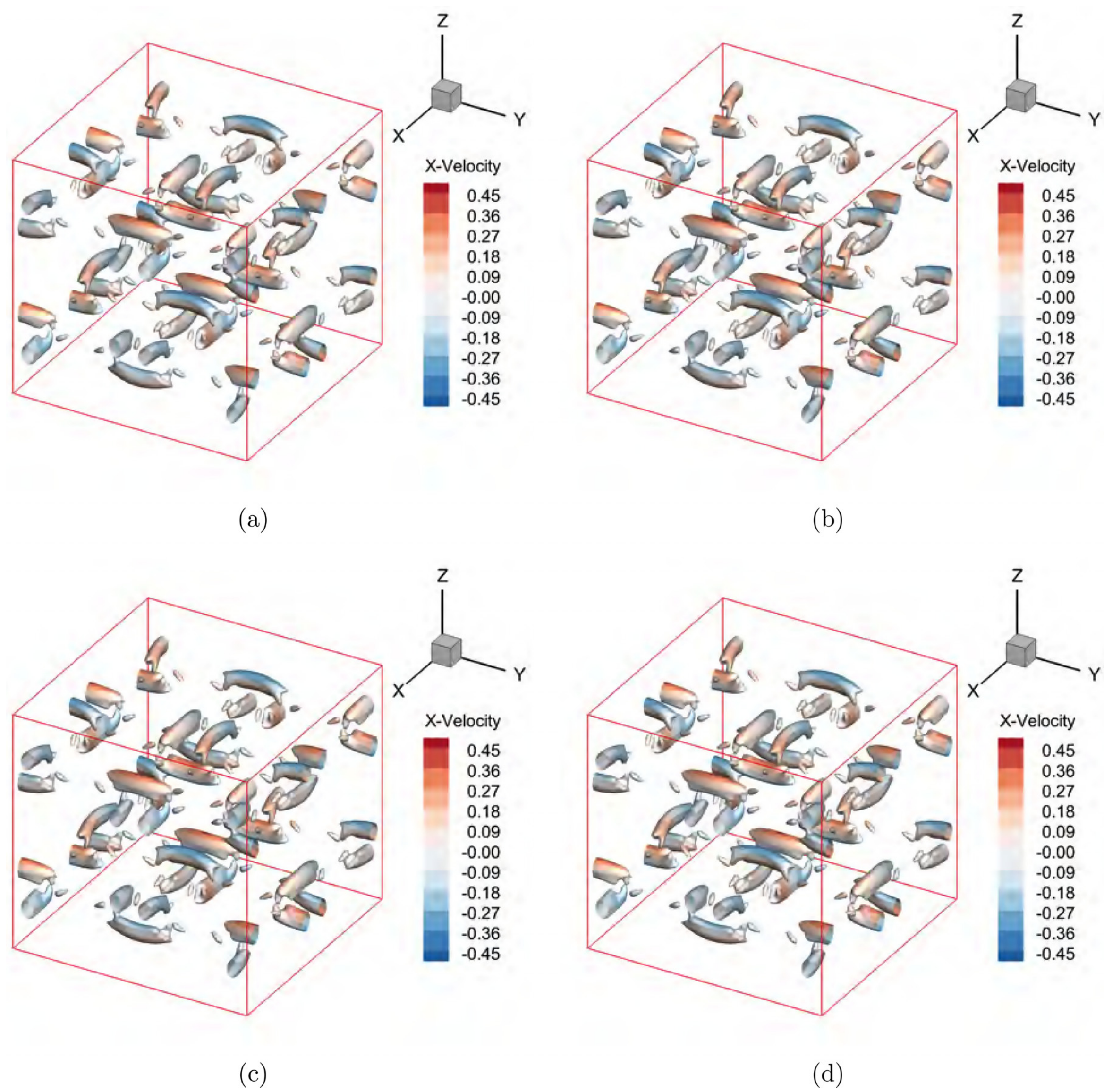


FIG. 15. *Q*-criterion ($Q = 1.0$ colored by the *x*-velocity) iso-surfaces of vortical structures obtained by different WENO-type schemes for viscous TGV problem with $Ma_{ref} = 2.5$. (a) WENO-JS, (b) US-WENO, (c) AUS-WENO, and (d) HAUS-WENO. $Re = 400$. $128 \times 128 \times 128$ grid points.

Reynolds number is 400, and the initial conditions are defined as Eq. (25). All boundary conditions are defined as a period, and the calculation time ends at $t = 10$.

In Fig. 13, the curves of kinetic energy [computed by Eq. (26)] over time for different WENO-type schemes with $128 \times 128 \times 128$ grid points are shown. The reference solutions are derived from the calculation results of the fourth-order scheme with $192 \times 192 \times 192$ grid points in Ref. 45. Unlike the case of $Re = 1600$, the differences between different WENO schemes seem to be less obvious, although all US-WENO schemes are slightly superior to the classical fifth-order WENO-JS scheme. This also means that for this case, the numerical dissipation of different WENO-type schemes has little effect on the simulation results. The intrinsic reason may be that the physical viscosity is smaller at $Re = 1600$, where the numerical viscosity of WENO schemes dominates in this situation. However, for smaller Reynolds numbers, the physical viscosity increases and is comparable to the numerical viscosity effect. The same conclusion can also be found in Figs. 14 and 15, which show the Q-criterion iso-surfaces calculated by different WENO schemes at the final moment. Due to the stronger compression effect, the vortex at $Ma_{ref} = 2.5$ appears to break more severely. In Figs. 16 and 17, the instantaneous velocity divergence surfaces (calculated by $DoV = u_x + v_y + w_z$) obtained by different WENO-type schemes at $t = 10$ are presented. The results show that the shocklet structures of US-WENO, AUS-WENO, and

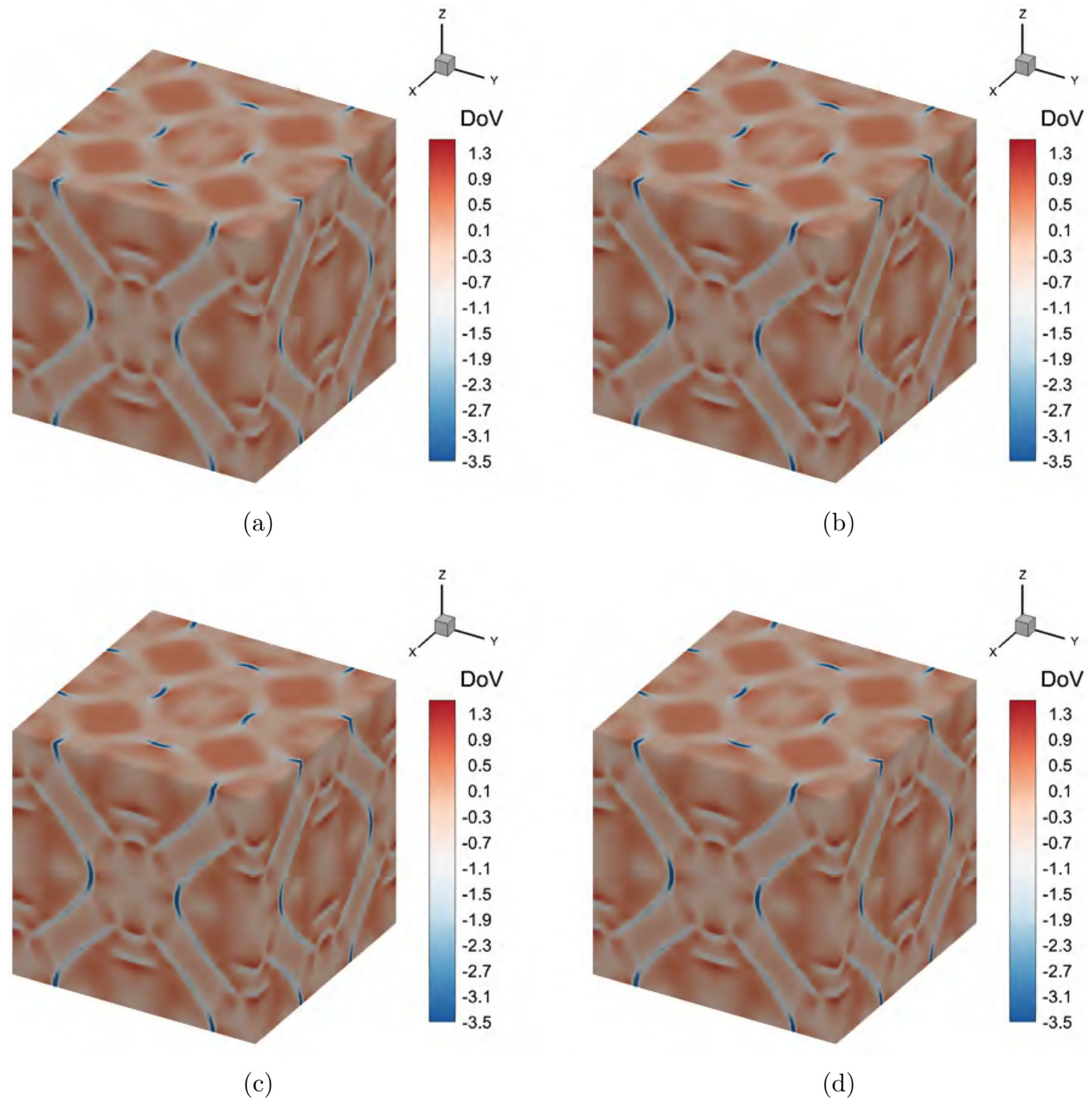


FIG. 16. The surfaces of velocity divergence obtained by different WENO-type schemes for viscous TGV problem with $Ma_{ref} = 2.0$. (a) WENO-JS, (b) US-WENO, (c) AUS-WENO, and (d) HAUS-WENO. $Re = 400$. $128 \times 128 \times 128$ grid points.

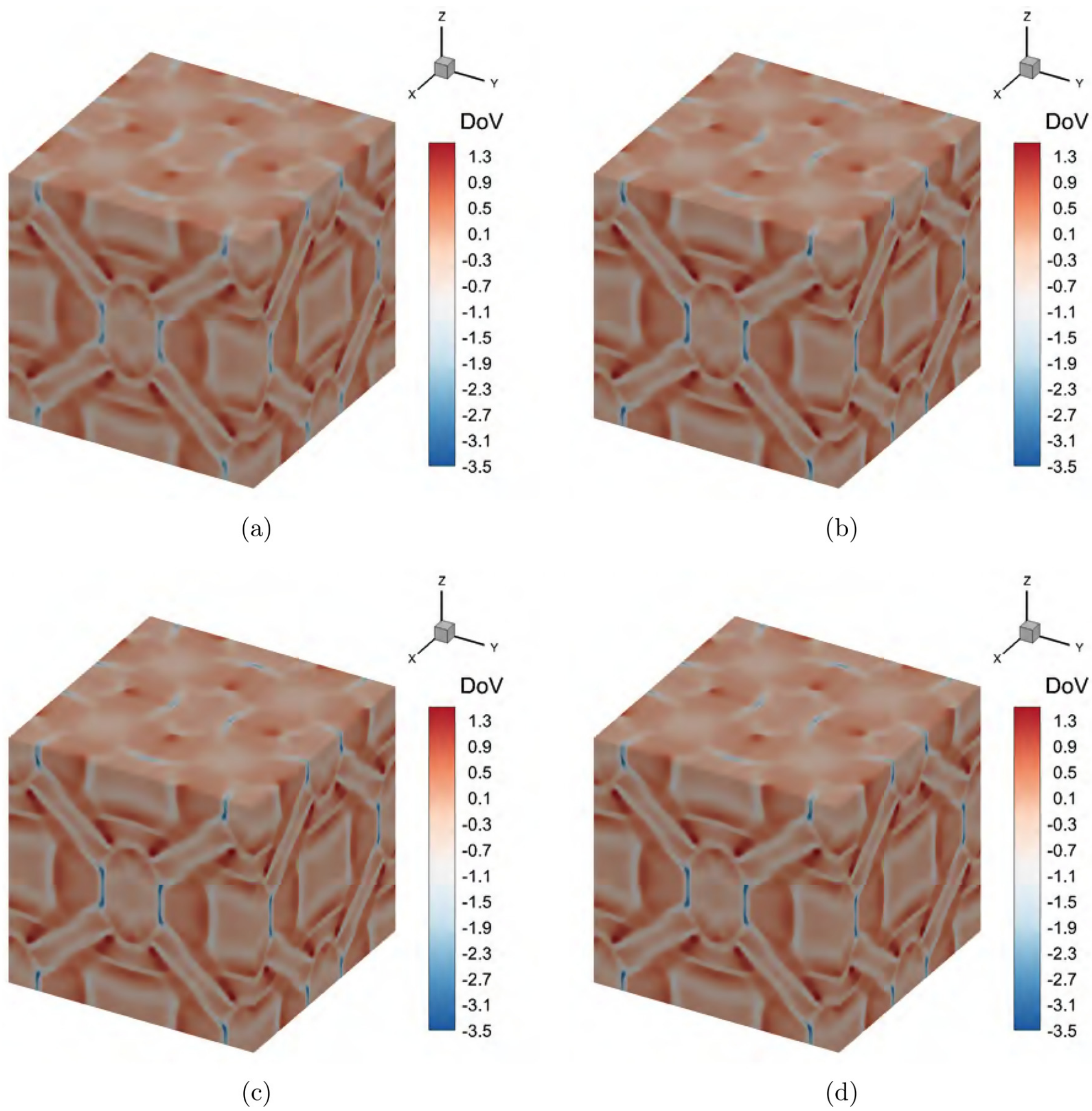


FIG. 17. The surfaces of velocity divergence obtained by different WENO-type schemes for viscous TGV problem with $Ma_{ref} = 2.5$. (a) WENO-JS, (b) US-WENO, (c) AUS-WENO, and (d) HAUS-WENO. $Re = 400$. $128 \times 128 \times 128$ grid points.

HAUS-WENO schemes are similar and that they are slightly sharper compared to the WENO-JS scheme.

Figures 18 and 19 show the dilatational component ε^D, solenoidal part ε^S, and kinetic energy total dissipation rate ε^T at Mach numbers 2.0 and 2.5, respectively. Different from kinetic energy, these quantities have greater discrimination for different WENO-type schemes. In the early stages of turbulence evolution, the dissipation of numerical methods has a relatively small impact on the results, while in the later stages, its impact is relatively greater. Overall, the interpolation-based AUS-WENO and HAUS-WENO schemes are closer to the reference solutions than the reconstruction-based US-WENO and WENO-JS schemes with the same mesh level.

E. Comparison of computational cost for different WENO schemes

In addition to numerical dissipation and shock wave capture capability, computational efficiency is also an important indicator that high-order methods need to consider. Table III shows the total and single-step computation time for all WENO schemes considered in this paper. Meanwhile, for ease of comparison, the calculation time of the WENO-JS scheme under the same conditions is used as a reference, and the relative ratios of other WENO schemes are provided. All simulations in this paper were performed on a single core with a 64 GB E5-2640V4 CPU. Note that in order to save space, only the results for $Ma_{ref} = 1.25$, $Re = 1600$ are given here.

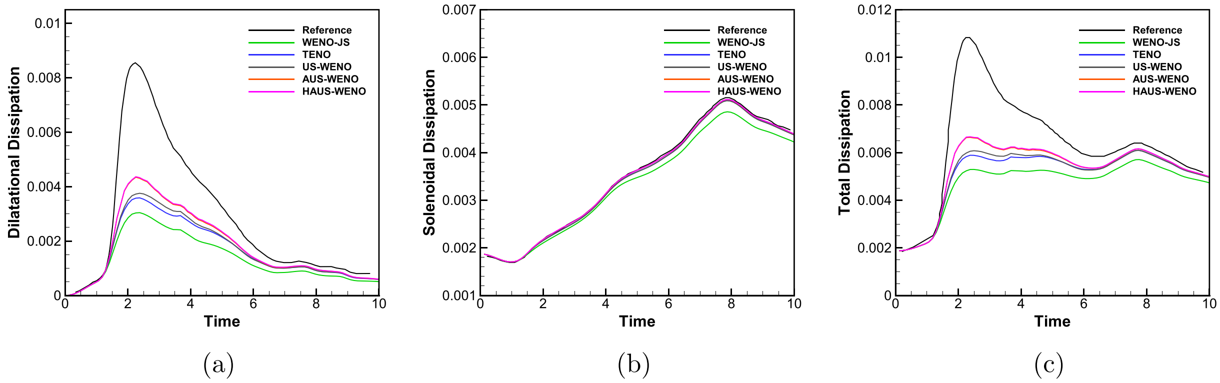


FIG. 18. Comparison of dilatational component, solenoidal part, and total dissipation rate vs time curves for viscous TGV problem with $Ma_{ref} = 2.0$. (a) Dilatational component ε^D, (b) solenoidal part ε^S, and (c) total dissipation ε^T. $Re = 400$. $128 \times 128 \times 128$ grid points.

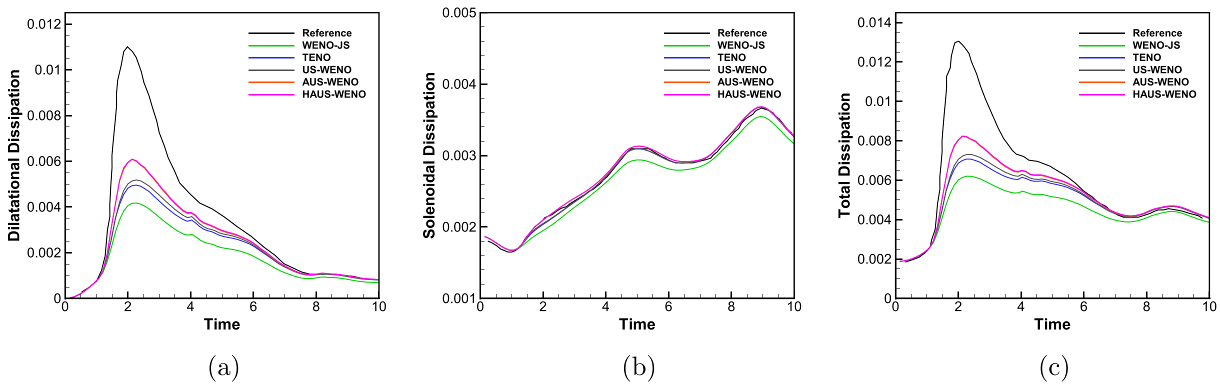


FIG. 19. Comparison of dilatational component, solenoidal part, and total dissipation rate vs time curves for viscous TGV problem with $Ma_{ref} = 2.5$. (a) Dilatational component ε^D, (b) solenoidal part ε^S, and (c) total dissipation ε^T. $Re = 400$. $128 \times 128 \times 128$ grid points.

TABLE III. Runtime and relative speed factor of the WENO-type schemes considered in this paper. $Ma_{ref} = 1.25$ and $Re = 1600$ case. $128 \times 128 \times 128$ grid points.

Scheme	Total runtime (s)	Ratio	Runtime of single step (s)	Ratio
WENO-JS	41 542.4	1.00	17.31	1.00
US-WENO	42 781.6	1.03	17.38	1.00
AUS-WENO	46 959.5	1.13	19.05	1.10
HAUS-WENO	30 046.8	0.74	12.09	0.70

From the table, it can be seen that the computational efficiency of the US-WENO and WENO-JS schemes is basically the same, only increased by 3%, while the AUS-WENO scheme is relatively more expensive, with an increase of 13%. The reason is that additional time is required to compute the high-order derivative terms in the interpolation-based spatial discretization process. The runtime of the HAUS-WENO scheme designed based on the improved discontinuous sensor developed in this paper is only 74% of that of the classical WENO-JS scheme, and the efficiency is improved by nearly 30%. Compared to the original AUS-WENO scheme under the same conditions, its efficiency has increased by 35%. Similar conclusions can also be drawn from comparing the runtime of single step. The reason behind this is that the HAUS-WENO scheme uses a cheaper linear method in smooth areas, which avoids the tedious process of local characteristic decompositions and nonlinear weight calculation. Combined with the previous conclusions, it is not difficult to see that the proposed HAUS-WENO scheme not only can inherit all the advantages of the original AUS-WENO scheme but also significantly improves computational efficiency and has better prospects.

IV. CONCLUDING REMARKS

In this paper, we evaluate and compare the performance of the reconstruction-based and interpolation-based US-WENO shock-capturing schemes on the typical shock/turbulence interaction problem of supersonic Taylor–Green vortex flows. The Mach numbers considered range from 0.1 to 2.5. On the whole, numerical results indicate that both reconstruction-based and interpolation-based US-WENO schemes can effectively reproduce complex physical phenomena such as transitions, fully developed turbulence, multiple shocklets, and shock-turbulence interactions. In terms of spatial accuracy, reconstruction-based and interpolation-based US-WENO schemes provided similar kinetic energy and large turbulence scale representations on relatively coarse grids. Overall, the reconstruction-based US-WENO scheme generates more small-scale dissipation at the same grid level, while the interpolation-based US-WENO scheme captures more small-scale structures. Regarding the dissipation rate and small-scale dynamics, greater differences were observed between reconstruction-based and interpolation-based US-WENO schemes, and the larger the Reynolds number, the more pronounced the differences. The reason lies in the relationship between numerical dissipation and physical viscosity. For larger Reynolds numbers, the physical viscosity is smaller and the difference between numerical dissipation increases, while as the Reynolds number decreases, the difference between the two decreases. In terms of computational efficiency, the interpolation-based US-WENO scheme generates some additional steps, which in turn makes it more time-consuming. However, the hybrid interpolation-based US-WENO scheme designed based on the EP-based sensor developed in this paper can inherit the low dissipation characteristics of the alternative US-WENO scheme while improving computational efficiency by about 30%. In the future, similar comparisons will be considered between interpolation-based and reconstruction-based WENO methods in more complex situations, such as shock wave boundary layer interactions.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Zhenming Wang: Conceptualization (equal); Formal analysis (equal); Methodology (equal); Project administration (equal); Validation (equal); Visualization (equal); Writing – original draft (equal). **Jun Zhu:** Funding acquisition (equal); Methodology (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal). **Linlin Tian:** Funding acquisition (equal); Methodology (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal). **Ning Zhao:** Funding acquisition (equal); Methodology (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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